

Automated, Accurate, and Scalable Relative Protein–Ligand Binding Free-Energy Calculations Using Lambda Dynamics

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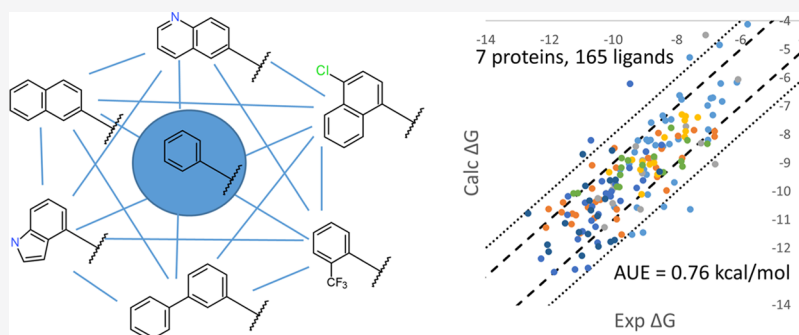
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ABSTRACT: Accurate predictions of changes to protein–ligand binding affinity in response to chemical modifications are of utility in small-molecule lead optimization. Relative free-energy perturbation (FEP) approaches are one of the most widely utilized for this goal but involve significant computational cost, thus limiting their application to small sets of compounds. Lambda dynamics, also rigorously based on the principles of statistical mechanics, provides a more efficient alternative. In this paper, we describe the development of a workflow to set up, execute, and analyze multisite lambda dynamics (MSLD) calculations run on GPUs with CHARMM implemented in BIOVIA Discovery Studio and Pipeline Pilot. The workflow establishes a framework for setting up simulation systems for exploratory screening of modifications to a lead compound, enabling the calculation of relative binding affinities of combinatorial libraries. To validate the workflow, a diverse data set of congeneric ligands for seven proteins with experimental binding affinity data is examined. A protocol to automatically tailor fit biasing potentials iteratively to flatten the free-energy landscape of any MSLD system is developed, which enhances sampling and allows for efficient estimation of free-energy differences. The protocol is first validated on a large number of ligand subsets that model diverse substituents, which shows accurate and reliable performance. The scalability of the workflow is also tested to screen more than 100 ligands modeled in a single system, which also resulted in accurate predictions. With a cumulative sampling time of 150 ns or less, the method results in average unsigned errors of under 1 kcal/mol in most cases for both small and large combinatorial libraries. For the multisite systems examined, the method is estimated to be more than an order of magnitude more efficient than contemporary FEP applications. The results thus demonstrate the utility of the presented MSLD workflow to efficiently screen combinatorial libraries and explore the chemical space around a lead compound and thus are of utility in lead optimization.

INTRODUCTION

Binding affinity of drug-like molecules to target proteins is a key property that is improved or monitored during the lead optimization stage of small molecule drug discovery. Lead optimization is a time- and cost-intensive stage,¹ which results from a large number of iterative cycles of design, chemical synthesis, and assaying of the compounds. Physics-based modeling methods can bring time and cost savings to lead optimization, by accurately predicting binding affinities of design compounds, because, unlike knowledge-based methods,² project and chemical series-specific training data are not required. Accurate predictions of binding affinities can reduce experimentation by prioritizing compounds likely to result in improved activities. Additionally, a larger chemical space may

be explored leading to higher quality compound designs that satisfy multiple property constraints simultaneously.

Explicit solvent-based free-energy methods offer a full atomistic description of protein–ligand systems within a physically rigorous framework, and thus, have the potential to accurately predict binding affinities. Within this class of methods, alchemical relative free-energy methods³ are

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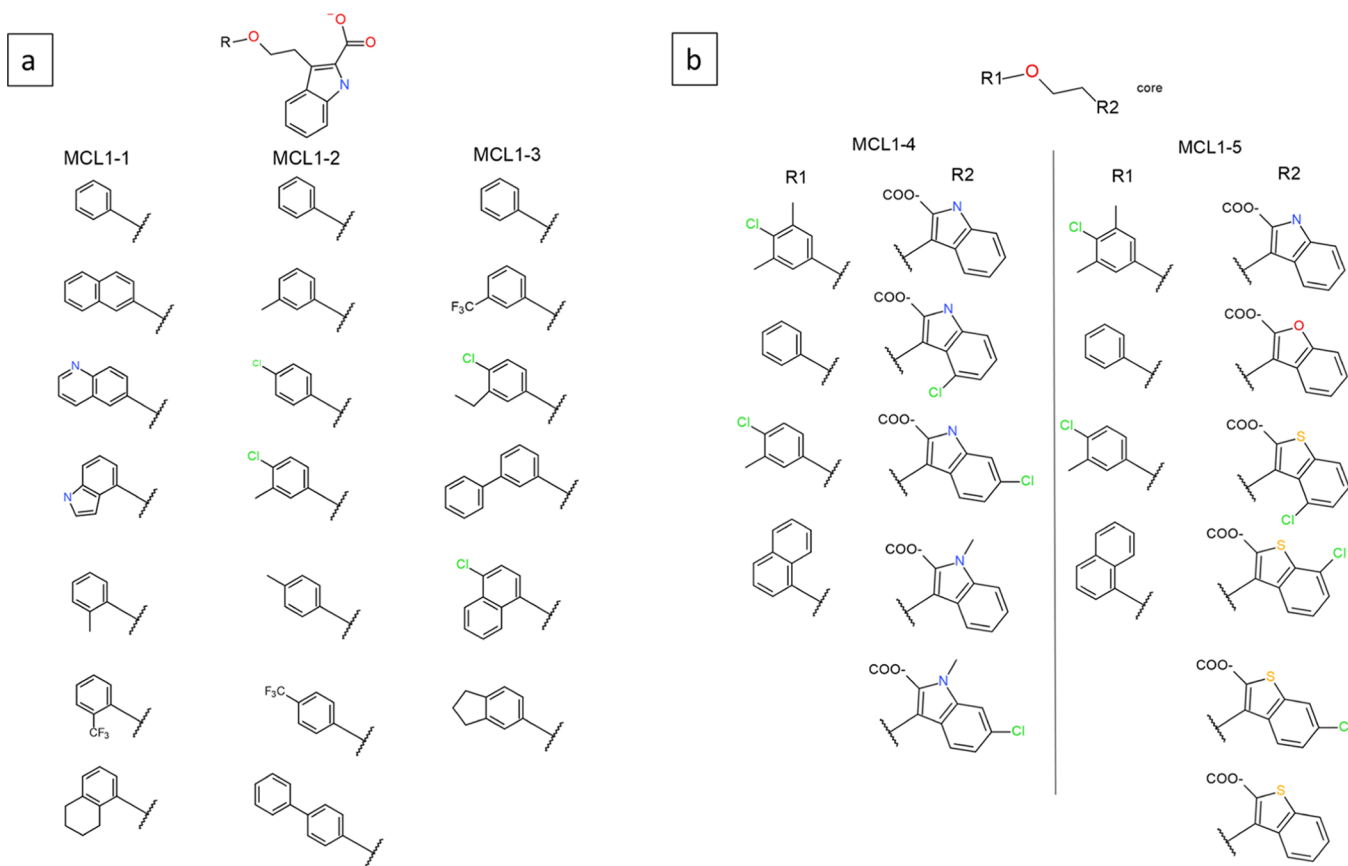


Figure 1. Five MSLD systems used to model ligands in the MCL1 data set. Panel (a) shows systems MCL1-1, MCL1-2, and MCL1-3. Panel (b) shows the two-site systems MCL1-4 and MCL1-5. The core ligand is shown on top.

particularly well suited for lead optimization as they require estimates of relative binding affinities in response to chemical changes to a lead compound. While relative free-energy methods are significantly more efficient than absolute free-energy methods, the computational cost is still quite high. Contemporary applications of relative free-energy perturbation (FEP) calculations typically required 60–390 ns of molecular dynamics sampling per ligand screened.^{4–6} Because this represents a significant investment in time and resources, there is a strong need to explore and develop methods that can accurately predict binding affinities at a lower computational cost.

Lambda dynamics (LD), introduced 24 years ago,⁷ is an alchemical relative free-energy method in which the coupling variable lambda is treated as a dynamic variable with a fictitious mass that is coupled to the system dynamics. In contrast to FEP or thermodynamic integration (TI), where a predetermined finite set of lambda values is sampled in multiple simulations, LD implements transformations between thermodynamic end states in a single simulation, with lambda evolution being driven by the intrinsic free-energy landscape. A simple probability-based estimator is used to predict the free-energy differences. Multiple ligands can be included, which for the problem of protein–ligand binding allows for the calculation of relative binding affinities of multiple ligands in a single LD simulation.^{8,9} To ensure efficient sampling of all end states, biasing potentials were initially introduced, which are subtracted during post-processing to recover the true free-energy differences. Over the years, the method has undergone enhancements including the extension of LD to multisite

lambda dynamics (MSLD)¹⁰ allowing for modeling substituents at distinct sites in a ligand, imposition of implicit constraints that bias the lambda sampling toward the end states,¹¹ and biasing potential replica exchange to enhance transitions between them.¹² Importantly, within the last few years, developments have been made including the introduction of additional biasing potentials^{13,14} to flatten the free-energy landscape of alchemical transformations, and a soft-core potential,¹³ which addressed prior limitations concerning sampling near end states. Another important methodological development involved an algorithm to optimize biasing potentials to promote free-energy landscape flattening^{13,14} which has been applied in recent retrospective studies.^{15,16}

The above-described developments in LD have set the stage for the application of this method in small molecule lead optimization. However, the following significant impediments exist in way of application to fast paced lead optimization projects. First, screening of multiple ligands in a single LD or MSLD simulation requires preparing a multitopology ligand that models a shared common core and separate substituents. Such a multitopology system is difficult to prepare and prone to errors when set up manually, even more so than for dual topology FEP systems. Second, while the adaptive landscape flattening (ALF) algorithm provides an automated way to tune parameters in the biasing potentials used in MSLD, the simulation lengths used in its iterations and other specifics of the algorithm have not been widely tested. Since the parameters in the biasing potentials need to be refined for each system, the protocol needs to be optimized by testing on a wide variety of protein–ligand systems to ensure effective

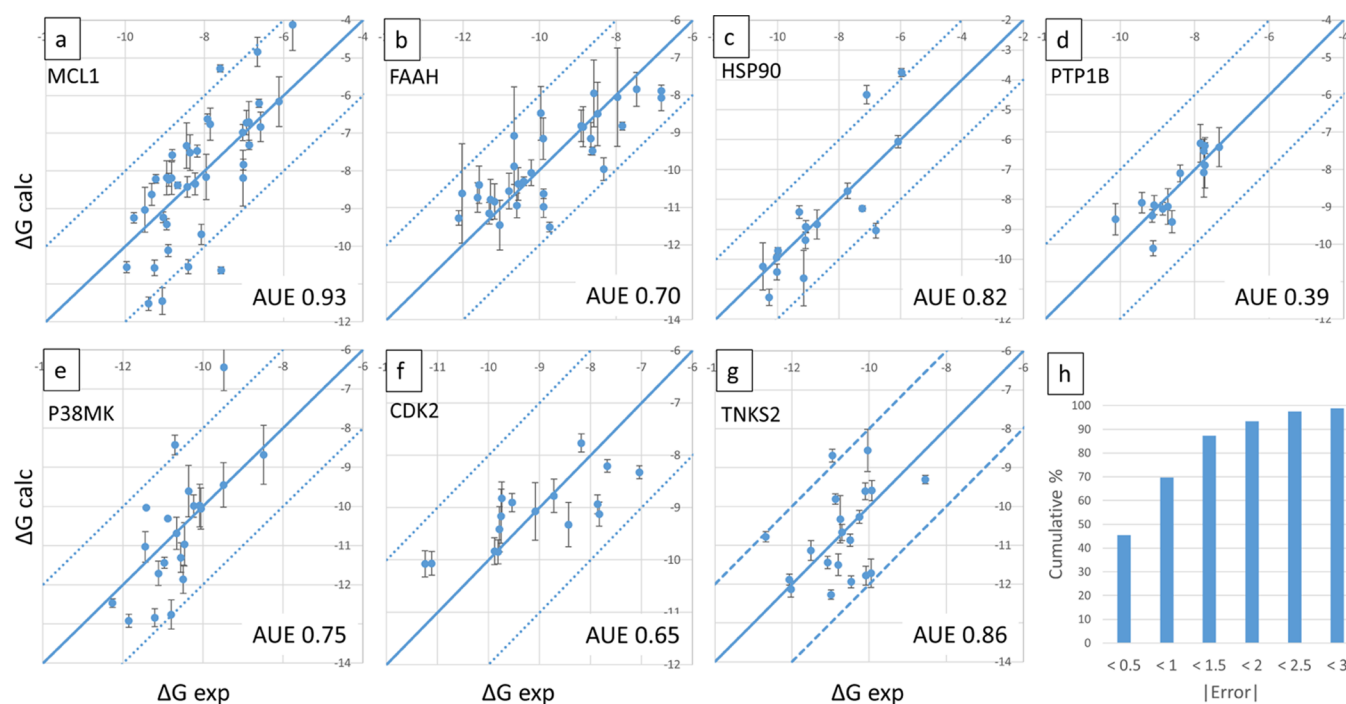


Figure 2. Calculated vs experimental absolute binding free energies for the seven data sets are shown in panels (a–g), with the name of the protein indicated on top left, and AUE on bottom right. Panel (h) shows the cumulative percentage of ligands for which absolute errors in predictions are lesser than different error bounds. Calculated relative free energies were converted to absolute by adding a constant offset value that minimizes errors associated with the choice of a single reference ligand, as described in the [Methods](#) section. Units are kcal/mol.

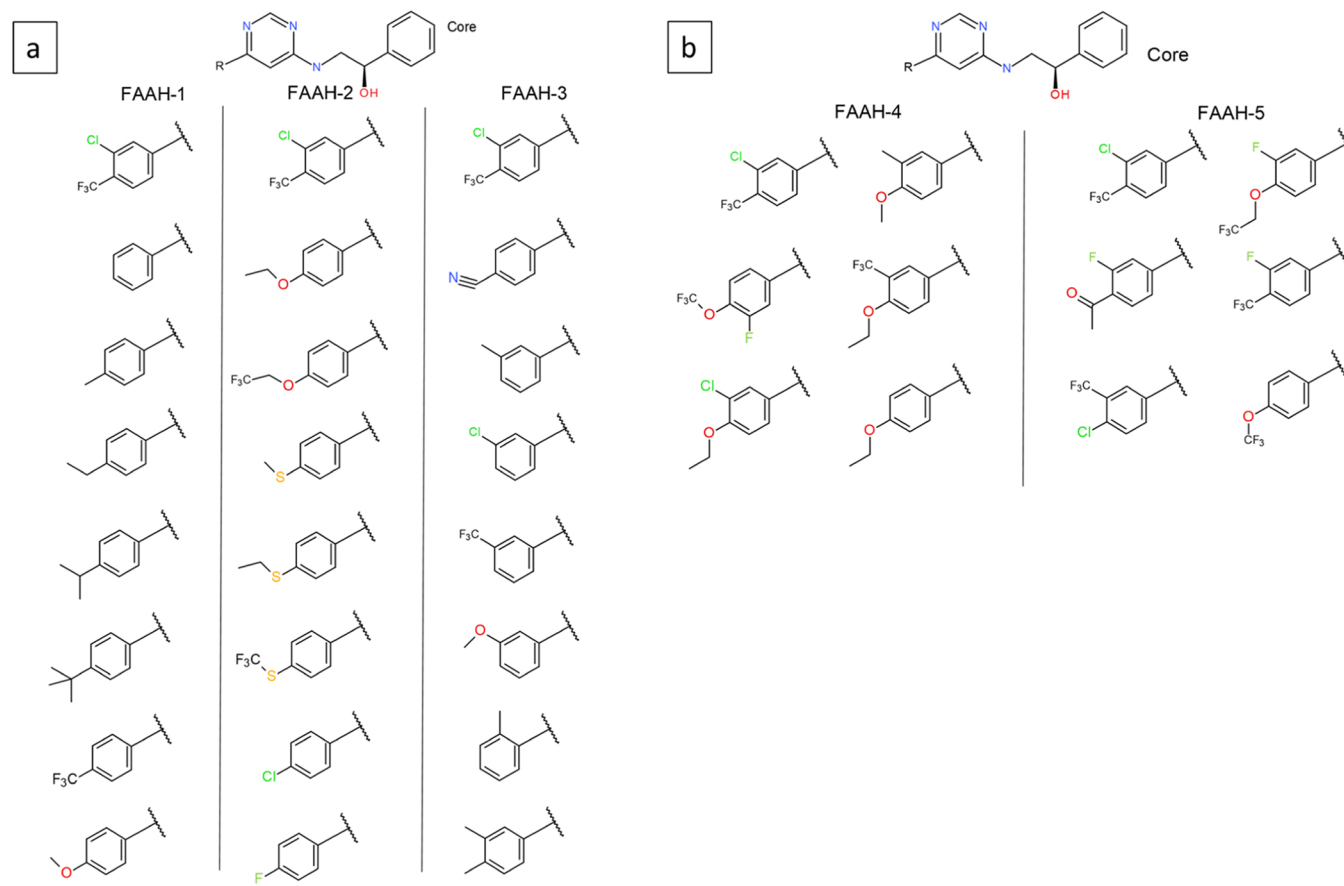


Figure 3. Five MSLD systems used to model ligands in the FAAH data set. Panel (a) shows systems FAAH-1, FAAH-2, and FAAH-3, which model eight substituents each. Panel (b) shows FAAH-4 and FAAH-5, which model six substituents each. The core ligand is shown on top.

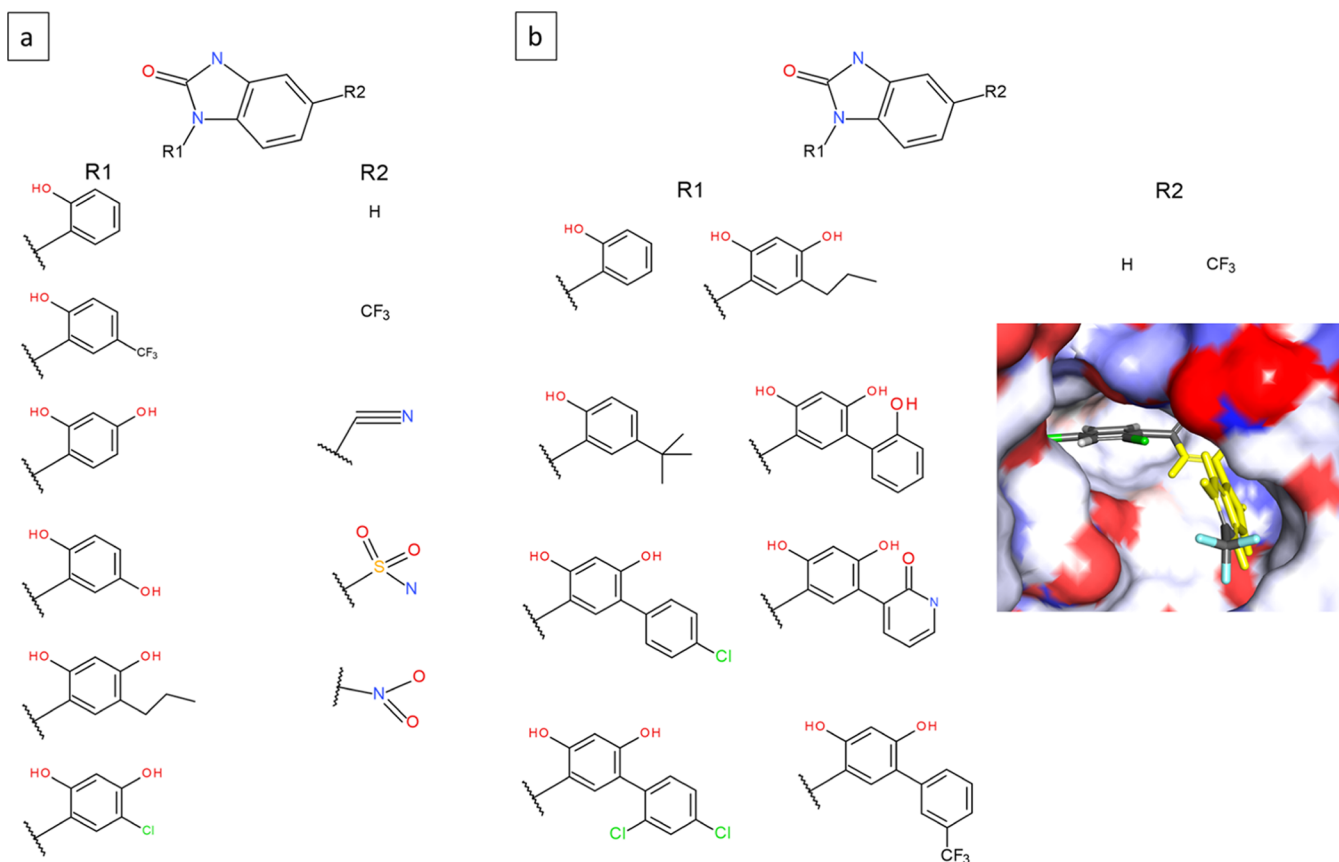


Figure 4. Two MSLD systems used to model ligands in the HSP90 data set. Panel (a) shows the HSP90-1 system with two sites. Panel (b) shows the HSP90-2 system also with two sites. Inset shows the partly occluded binding pocket near site 1. The core ligand is shown on top.

flattening and to identify minimal simulation lengths for obtaining accurate results. Finally, other system setup requirements including partial charges, force-field parameter assignment for the ligands and proteins, assignment of initial coordinates for the ligands, and generation of solvated simulation systems also need to be reached for MSLD calculations, as with FEP calculations. This paper is aimed at addressing these needs by developing an automated workflow for setting up and executing MSLD calculations and retrospectively validating it.

The automated MSLD workflow implemented in BIOVIA Discovery Studio and Pipeline Pilot packages¹⁷ is described in detail in the [Methods](#) section. Given a core ligand, multiple sites of modification, and the specification of modifications or substituents, the workflow provides tools to set up and execute the MSLD calculations to estimate the binding free energies of a full combinatorial library. A related consideration in lead optimization in addition to screening efficiency is coverage of a broad chemical space in relation to the lead modifications. Keeping this consideration in mind, the MSLD workflow was designed to support the modeling of chemical modifications, which include not only substituents (hydrogen atom replacements), but more broadly, arbitrary modifications, as shown in [Figures 1–8](#). The validation data set was chosen appropriately to represent both small and large lead modifications on a common core. Validation results suggest that the presented MSLD workflow can accurately calculate binding affinities of large combinatorial libraries with significantly greater efficiency than FEP, and thus be of utility in rapidly exploring chemical

space to optimize binding affinity, or monitoring it while other drug-like properties are optimized in concert.

METHODS

Lambda Dynamics. The λ -dynamics methodology has been described in detail in previous works.^{7,8} Briefly, in contrast with FEP and TI calculations, λ is treated as a dynamic variable. The dynamics of the system is generated using an extended Hamiltonian

$$H_{\text{extended}}(X, \{\lambda\}, \{x\}) = T_x + T_\lambda + V(X, \{\lambda\}, \{x\}) + V_{\text{bias}}(\{\lambda\})$$

where T_x and T_λ represent the kinetic energies of the atomic coordinates and λ variables. Because λ is treated as a dynamic variable, all intermediate states may be sampled during a single simulation. The third term represents the hybrid potential energy function for the solvent, the protein if present, and the alchemical ligand, and the fourth term is a bias described later. For a single perturbation site with a total of N ligands, the hybrid potential term is given by

$$V(X, \{\lambda\}, \{x\}) = \sum_{i=1}^N \lambda_i V(X, x_{S,i}) + V_{\text{env}}(X)$$

where X and x_i are the coordinates of the environment and ligand core atoms and of the ligand substituent atoms, respectively, as described below. λ_i is the coupling parameter associated with ligand i . MSLD¹⁰ was developed to enable the

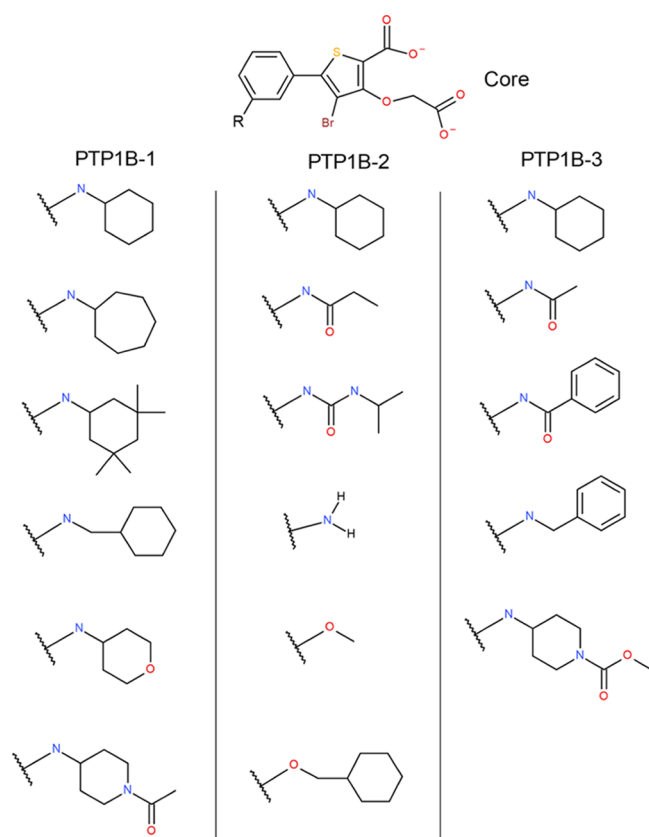


Figure 5. Three MSLD systems used to model ligands in the PTP1B data set. The core ligand is shown on top.

use of multiple substituents at multiple sites, and the hybrid potential energy function was extended.

$$V(X, \{\lambda\}, \{x\}) = V_{\text{env}}(X) + \sum_{S=1}^{M_{\text{sites}}} \sum_{i=1}^{N_S} \lambda_{S,i} V(X, x_{S,i}) \\ + \sum_{S=1}^{M_{\text{sites}}-1} \sum_{i=1}^{N_S} \sum_{T=S+1}^{M_{\text{sites}}} \sum_{j=1}^{N_T} \lambda_{S,i} \lambda_{T,j} (V(x_{S,i}, x_{T,j}))$$

M_{sites} is the total number of sites that contain multiple substituents, while N_S is the number of substituents at site S off the common ligand core. The double summation in the second term accounts for the interactions between the environment and each substituent at each site. The third term accounts for interactions between each substituent and the substituents modeled at all other sites. However, it is important to note that at a given site, substituents do not interact or “see” one another during the simulations. For single-site λ -dynamics, a ligand is dominant when its λ value is greater than $\lambda_{\text{cut}} = 0.99$. For MSLD, a ligand is described as dominant when the λ values associated with its constituent substituents are all dominant at the same time. Within MSLD, constraints are maintained at each site α according to

$$0 \leq \lambda_{\alpha,i} \leq 1 \text{ and } \sum_{i=1}^{N_{\alpha}} \lambda_{\alpha,i} = 1$$

To satisfy these constraints, we use an implicit constraint approach¹¹ and simply define λ 's for the N substituents at site α to be

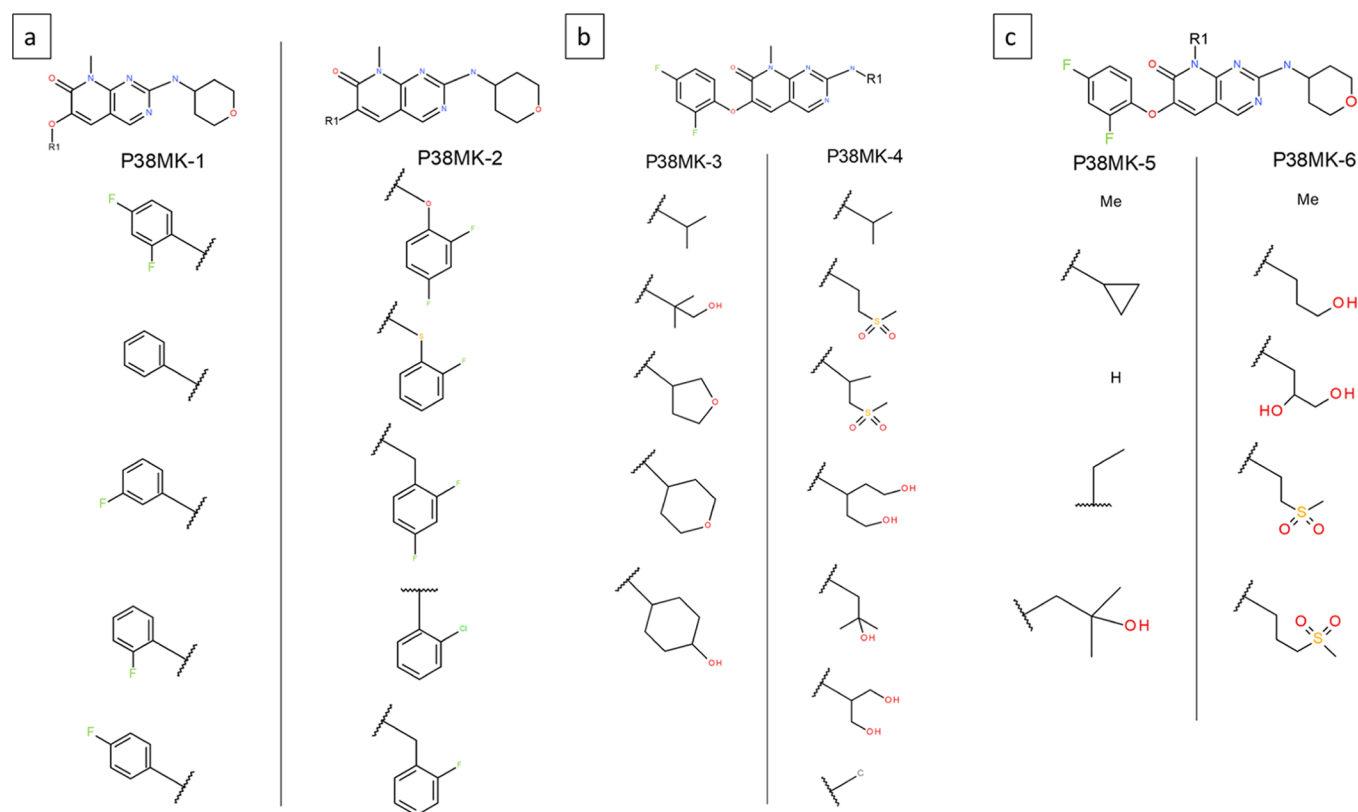


Figure 6. Six MSLD systems used to model ligands in the P38 MAP kinase data set. Panel (a) shows systems P38MK-1 and 2. Panel (b) shows systems P38MK-3 and 4. Panel (c) shows systems P38MK-5 and 6. The core ligand is shown on top.

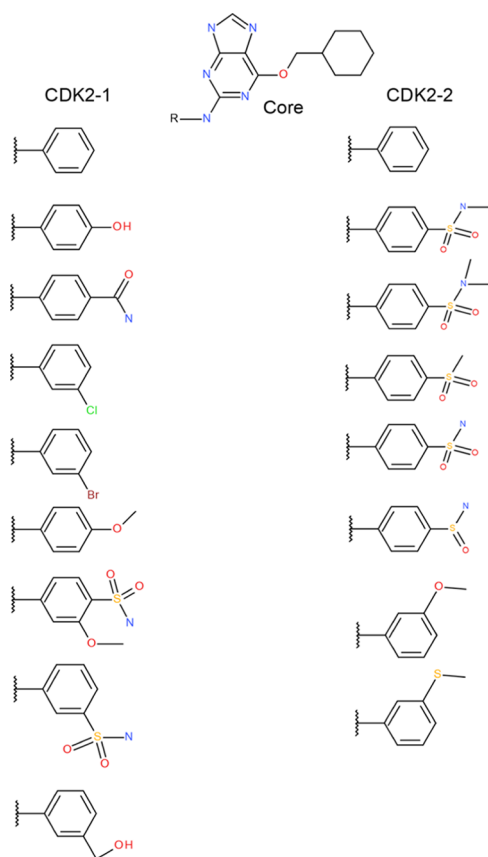


Figure 7. Two MSLD systems used to model ligands in the CDK2 data set. The core ligand is shown on top.

$$\lambda_{\alpha,i} = e^{5.5 \sin \theta_{\alpha,i}} / \sum_{j=1}^{N_{\alpha}} e^{5.5 \sin \theta_{\alpha,j}}$$

Therefore, for MSLD, it is values of θ that have fictitious masses, m_{θ} , and velocities, $\dot{\theta}$ and are propagated through the equation of motion and transformed back to their corresponding λ values used in free-energy determination. The alchemical kinetic energy term is thus defined as

$$T_{\lambda} = \sum_{S=1}^{M_{\text{sites}}} \sum_{i=1}^{N_S} \frac{1}{2} m_{\theta} \dot{\theta}^2$$

Protein Structure Preparation. Protein structures corresponding to the congeneric ligand sets were downloaded from the Protein Data Bank (PDB).¹⁸ Only one protein structure per congeneric set was used, as listed in Table 1. Protein structures were prepared using the “prepare protein” protocol in BIOVIA Discovery Studio. The p38 MAP kinase protein structure involved two residues with missing coordinates, which were automatically built by the protocol. The protein preparation protocol involves generalized Born electrostatics-based assignment of protonation states, consistent with a pH of 7.4, assuming a protein dielectric constant of 10 and ionic strength of 0.145 M, as detailed previously.¹⁹ Any water molecules present in the crystal structures were retained. The CHARMM36²⁰ force-field was used to model the proteins.

Ligand Preparation and Binding Pose Generation. Ligands obtained from various sources were either in SMILES format or were converted to the same. Ionization states of

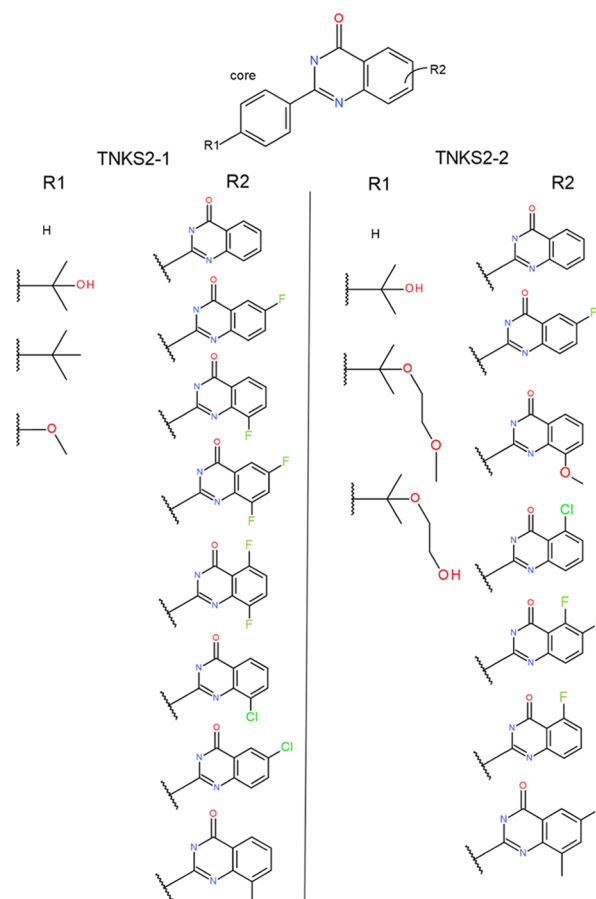


Figure 8. Two MSLD systems used to model ligands in the TNKS2 data set. Panel (a) shows the TNKS2-1 system with two sites. Panel (b) shows the TNKS2-2 system also with two sites. The core ligand is shown on top.

Table 1. Data Set Used for Retrospective Validation^a

protein	PDB ID	<i>n</i> ligands	range	AUE	RMSE	<i>R</i>
MCL1 ^b	4HW3	39	4.19	0.93	1.19	0.73
FAAH ^c	6MRG	32	5.30	0.70	0.87	0.80
HSP90 ^d	4YKR	16	4.50	0.82	1.18	0.83
PTP1B ^b	2QBS	15	2.78	0.39	0.48	0.84
P38MK ^b	3FLY	27	3.80	0.75	1.07	0.71
CDK2 ^b	1H1Q	16	4.21	0.65	0.79	0.78
TNKS2 ^e	4UI5	20	4.13	0.86	1.09	0.43

^aThe protein target, PDB ID of the protein structure used, the number of ligands, and experimental affinity range of each set are listed. Also, listed are the prediction accuracy metrics from the calculations presented here, the AUE, RMSE, and correlation coefficient (*R*) for each set. Units of range, AUE, and RMSE are kcal/mol. ^bData source: Wang et al. ^cSaha et al. ^dD3R. ^eSchindler et al.³³

ligands were retained as existing in the publication or data source except for HSP90 ligands, where they were manually inspected. For the HSP90 set, a “rule-based” method was used to assign a single ionization state for the ligands, following which 2D coordinates were generated. The 3D geometry or binding pose for the congeneric series of ligands in each data set was assigned using the recently validated “generate analog conformations” (GAC) method.²¹ This method uses the experimental structure of protein in the complex with one ligand in the series as a template, which in this study happens

to be the PDB structure listed in Table 1. The method detects the maximal common substructure (MCS)²² between the template ligand and other ligands in the data set, from which it ascertains the common set of atoms, and atoms that are different. For each ligand, it then directly assigns coordinates for the common set of atoms from the template ligand. For the remaining atoms, a restricted conformational generation procedure is implemented that generates multiple conformations. These conformations are scored using the Molecular Mechanics-Generalized Born with Solvent Accessible Surface Area (MM-GBSA) method²³ as a first step, and then, the protein–ligand complex is energy minimized with non-local (>3 Å away) protein residues held rigid. Following the minimization, a single point MM-GBSA energy calculation is performed to calculate a binding score, which is used to rank the different geometries. The top scoring geometry is selected for the MSLD calculation setup. The generalized Born molecular volume method was used to calculate the solvation energy contribution, which used dielectric constants of 1 and 80 for the solute and solvent, respectively. In addition, a linear solvent accessible surface area term with slope (α) of 0.00542 kcal mol⁻¹ Å⁻² and intercept (β) of 0.92 kcal mol⁻¹ was included. Further details of the GAC method are available in our previously published study.²¹ Force-field parameters and partial charges for ligands were assigned, consistent with CGenFF²⁴ using the MATCH algorithm²⁵ included in Discovery Studio and Pipeline Pilot. For HSP90, P38MK, CDK2, and TNKS2 set ligands, a new atom type implemented in the developmental version of the software was used.

Multitopology Builder and Simulation System Setup.

A MSLD topology builder was implemented that takes as input one core ligand, and multiple additional ligands, each representing a different substituent corresponding to a modification at one of the sites. Each non-core ligand is expected to model one chemical modification with respect to the core at a single site only and carries a property designating the identity (index) of that site. Ligands corresponding to each site are grouped together, and for each group belonging to a site, a MCS is calculated with the core ligand. Details of the MCS algorithm are described elsewhere.²² When multiple MCS solutions are returned, the one chosen is the one that results in the minimum root mean squared deviation (rmsd) based on the positions of the constituent atoms in the ligands. Hydrogen atoms are not considered in MCS computation for algorithmic efficiency and are subsequently added to the MCS based on the mapping of the parent heavy atoms. The resultant MCS is further reduced by removing atoms that have different force-field atom types and/or partial charge assignment. Hydrogen atoms attached to heavy atoms identified during this step are also removed. In this way, an atom–atom mapping is obtained between the core ligand and all ligands, representing a substituent at one site. This process is repeated for all M sites separately, resulting in M MCS solutions. Following this, an intersection of the M MCS is calculated which results in a single MCS. The MCS contains an atom–atom mapping, which encodes the identities of the common core atoms and atoms that belong to different substituents at different sites. A multitopology hybrid molecule is created which contains a single copy of the core atoms and one copy of atoms for each substituent in the different sites. The force-field atom types, partial charges, coordinates, bond connectivity, and improper dihedral definitions for the hybrid molecule are copied from the constituent ligands. Care is taken not to

duplicate the bond and improper dihedral definitions involving MCS atoms. Following the creation of the multitopology ligand, the ligand is solvated in a water box, and a copy is merged with the protein structure and solvated. The solvation step includes neutralizing Na⁺ and Cl⁻ ions, consistent with an ionic strength of 0.145 M. Cubic boxes are used to encapsulate the protein–ligand complex and the free ligand systems and ensure at least 7 Å between the extremities of the solute and the edge of the box. Two solvated systems are used in MSLD calculations. For a few protein–ligand complex systems simulated in this study, orthorhombic boxes were used instead of cubic to reduce system size. This setup feature was introduced at a later stage during the development of the method and thus only a few systems were modeled with this approach, which include CDK2 systems and P38MK-8 systems. Proteins, that may have an elongated shape, when incorporated into the orthorhombic boxes have the potential to rotate during the course of the simulation and consequently come in contact with its own periodic image. This problem does not exist for cubic systems due to all edge lengths being the same. When using orthorhombic boxes, weak restraints with a force constant of 0.01 kcal mol⁻¹ Å⁻² were applied to the C α carbon atoms of the protein to restrain the translation and rotation of the protein to avoid interaction with its periodic image. Restraints were not applied to binding site residues automatically chosen, based on a 5 Å proximity criterion to the ligand. A protocol named “set up MSLD calculations” was implemented in Discovery Studio and Pipeline Pilot that generates the MSLD simulation systems given input ligands. Additionally, a protocol named “enumerate ligands for MSLD” was developed that makes it easier to generate input ligands for the above protocol. This protocol requires a core ligand and the identification of the sites of modification and the R-group attachments in SMARTS format.

Bias Optimization. MSLD simulations use biasing potentials to facilitate transitions among the substituents at each site. Four biasing potentials are used in this implementation, as detailed previously,^{13,14} namely fixed, quadratic, end, and skew biases.

$$V_{\text{bias}} = V_{\text{Fixed}} + V_{\text{Quad}} + V_{\text{End}} + V_{\text{Skew}}$$

The functional forms of the lambda-dependent potentials are described below.

$$V_{\text{Fixed}} = \sum_s^M \sum_i^{N_s} \phi_{si} \lambda_{si}$$

$$V_{\text{Quad}} = \sum_s^M \sum_i^{N_s} \sum_{j>i}^{N_s} \psi_{si,sj} \lambda_{si} \lambda_{sj}$$

$$V_{\text{End}} = \sum_s^M \sum_i^{N_s} \sum_{j \neq i}^{N_s} \omega_{si,sj} \lambda_{si} \lambda_{sj} / (\lambda_{si} + \alpha)$$

$$V_{\text{Skew}} = \sum_s^M \sum_i^{N_s} \sum_{j \neq i}^{N_s} \chi_{si,sj} \lambda_{sj} (1 - \exp(-\lambda_{si}/\sigma))$$

where $\alpha = 0.017$ and $\sigma = 0.18$ were previously optimized to yield good fits to free-energy profiles for a variety of systems.^{14–16} The set of constants, $\{\phi_{si}\}$, $\{\psi_{si,sj}\}$, $\{\omega_{si,sj}\}$, and $\{\chi_{si,sj}\}$ are optimized during bias optimization.

The bias optimization algorithm is iterative in nature where short MSLD simulations are run, followed by calculation of free-energy landscapes. Three types of landscapes are computed, the first one being a function of a single lambda value, and the remaining two being a function pairs of lambda values. Single lambda landscapes, $G(\lambda_{si})$, are calculated for each substituent on a site. Free-energy landscapes corresponding to transitions between a pair of substituents on a site, $G(\lambda_{si}/(\lambda_{si} + \lambda_{sj}))|_{\lambda_{si} + \lambda_{sj} > 0.8}$, are calculated by using data restricted to the lambda space corresponding to transitions between the two substituents. Pairwise landscapes, $G(\lambda_{si}\lambda_{sj})$, are also calculated using unrestricted data for all pairs of substituents on a site. In a multitopology ligand, there are N_s $G(\lambda_{si})$ landscapes and $N_s(N_s - 1)/2$ pairwise landscapes for each site s . $G(\lambda_{si})$ and $G(\lambda_{si}/(\lambda_{si} + \lambda_{sj}))|_{\lambda_{si} + \lambda_{sj} > 0.8}$ landscapes are calculated by binning the lambda values obtained from the trajectory into 400 bins, and $G(\lambda_{si}\lambda_{sj})$ are calculated by binning pairs of lambda values into 20×20 bins. Implicit constraints¹¹ are present in LD simulations, which are subtracted from the landscapes to avoid flattening the entropic barrier of about 3 kcal/mol in the mid-lambda range. The entropic barrier is retained since it is a desirable feature that promotes more sampling in the end states. A running window average of the landscapes is calculated using the most recent six simulations, where thermodynamic averaging is performed using the weighted histogram analysis method.¹³

For each landscape, a flatness parameter is calculated as the rmsd from the $F = 0$ line, and an average flatness is calculated over all single and pair lambda landscapes, respectively. For pairs of substituents, the flatness parameter is only calculated for the transition landscapes. Iterations are terminated when the average flatness falls below a specified threshold value. This is done so as to avoid running further simulations if the free-energy landscapes are sufficiently flat. However, this termination criterion is only invoked in phase 1 of the bias optimization protocol (see below). One set of biases is chosen from the last six iterations that provides the most even sampling for all substituents on all sites, which is implemented by maximizing an entropy metric calculated as follows

$$S = \sum_s \sum_i^{N_i} -p_{si} \ln p_{si}$$

where p_{si} is the probability of $\lambda_{si} > \lambda_{\text{cut}}$ (and $\lambda_{\text{cut}} = 0.99$).

A three phase bias optimization protocol was implemented. The first phase runs 100 iterations of 100 ps long MSLD simulations. If the average flatness parameters associated with single and pair (transition) lambda landscapes described above reach a value below 0.25 and 0.5 kcal/mol, respectively, iterations are terminated. The second phase of bias optimization runs 10 iterations of 1 ns long MSLD simulations, and the third phase runs five iterations of 2 ns long simulations. At the end of each phase, the last six iterations are evaluated, and the biases associated with the simulation that results in the maximum entropy metric are chosen for the next phase or for production simulations.

MSLD Production Simulations. All simulations were performed under periodic boundary conditions. The van der Waals interactions were switched off smoothly in the range 10–12 Å, and the particle mesh Ewald method^{26,27} was used to treat long-range electrostatics with a real space cut off of 12 Å. An integration time step of 2 fs was used for dynamics. Production simulations were performed in the *NPT* ensemble,

where a temperature of 298.15 K was maintained using the Nose–Hoover thermostat method and pressure of 1 atm was maintained using the Langevin piston method. A soft-core potential¹³ was used to alleviate end-point singularities. The MD simulation parameters used in the production stage are identical to those used in the bias optimization stage.

Production runs involve six independent trajectories, which are each 20 ns long each, for a cumulative sampling time of 120 ns. Two series of production runs are performed, one for the free ligand in solvent and one for the solvated protein–ligand complex. From each trajectory, the free energy of each combinatorial ligand c from trajectory t is calculated as follows

$$F_c^t = -k_B T \ln p_c^t - V_{\text{Fixed},c}$$

where p_c^t is the probability of all substituents involved in combination c to have their respective lambda values $\lambda_{si} > \lambda_{\text{cut}}$. $V_{\text{Fixed},c}$ is the sum of fixed biases corresponding to the substituents in combination c . The fixed bias is subtracted to retrieve the unbiased true free energy. Estimates from trajectories 1–2, 3–4, and 5–6 are combined to obtain three free energy values for each combination c . A Boltzmann average of free-energy estimates from two trajectories ti , tj is obtained as follows.

$$F_c^{ti,tj} = -k_B T \ln \frac{\exp(-F_c^{ti}/k_B T) + \exp(-F_c^{tj}/k_B T)}{2}$$

The free energy associated with a combination F_c is calculated as a simple average over $F_c^{t1,t2}$, $F_c^{t3,t4}$, and $F_c^{t5,t6}$ values—a corresponding standard deviation is also calculated. Relative binding free energies are calculated by taking of the difference of F_c calculated in the complex and free ligand states. A standard deviation associated with the prediction is calculated as the square root of the sum of squares of the standard deviations associated with complex and free ligand states.

From each MSLD simulation, the time series of lambda values corresponding to each substituent output at a frequency of 0.02 ps was extracted. The lambda time series was transformed into a “state time series” which indicates the identity of a substituent si that is active at a given time point $\lambda_{si} > \lambda_{\text{cut}}$. The state time series is processed individually for each site. For each substituent si , the number of times the system transitions from any substituent sj to si , or from si to any substituent sj is counted as the number of transitions associated with si . Transition rates are calculated for pairs of trajectories over a total sampling time of 40 ns, resulting in three estimates which are averaged to obtain the expected number of transitions over 40 ns.

Comparison to Experiment and Prediction Metrics. Validation included one-site and two-site systems that modeled 5–32 ligands per system, and two- and three-site systems that modeled large combinatorial libraries of 64–180 ligands per system. Twenty-five one-site and two-site systems were created by dividing the seven data sets into smaller subsets. For a given protein, subsets were created so as to maintain a common reference ligand in all sets to enable the calculation of absolute binding affinities by using the experimental affinity of the one reference ligand. The prediction metric used to assess accuracy in the 25 small subsets was average unsigned errors (AUE) of $\Delta\Delta G$ values with respect to the corresponding experimental values. To observe the overall prediction quality for each protein set, the absolute binding affinity ΔG was calculated by

first using the experimental ΔG of one common ligand used as a reference in each subset. Following the approach in other retrospective free-energy studies,^{4,5} a single offset value was added to the calculated ΔG value to minimize errors associated with the choice of a single reference ligand as the lead compound with known affinity. The same approach was followed for the calculation of the AUE of ΔG for the large-scale multisite systems modeled in this study. Other metrics including root mean square error (RMSE), and correlation coefficient (R) were also calculated to assess the quality of predictions for each protein set.

MSLD Workflow. The MSLD workflow implemented in the Discovery Studio and Pipeline Pilot packages¹⁷ was used to run all calculations. The 2020 release of the software included prototype MSLD functionality that incorporates protocols to set up MSLD systems, run bias optimization, run production, and collate results. In addition, a helper protocol for enumerating R-group features on a core ligand and another protocol to generate initial binding poses²¹ of ligands are also included. Calculations discussed in this study were run with a developmental version of the software that added a protocol to combine bias optimization, and production to support multiphase bias optimization, with enhanced efficiency and usability. The MSLD simulations during bias optimization and production stages are run with CHARMM²⁸ included in the software using the DOMDEC-GPU platform.²⁹ These stages can be run on a Pipeline Pilot server installed on a GPU cluster (grid) with a supported queuing system to allow for automatic queuing and execution.

RESULTS

To evaluate the accuracy and reliability of the MSLD workflow, it was tested on a diverse data set, which is described below. All steps of the workflow are implemented in Discovery Studio and Pipeline Pilot packages,¹⁷ which include protein preparation, ligand-binding pose generation, force-field assignment, MSLD multitopology setup, MSLD bias optimization, production, and analysis. These steps are described in detail in the [Methods](#) section. For each MSLD system prepared, a three phase bias optimization was performed with up to 10 ns of simulation time in each phase, for a cumulative time of 30 ns. The number of iterations and iteration length in the three phases were 100×100 ps, 10×1 ns, and 5×2 ns, respectively. Phase 1 is initiated with all biases set to zero, and short iterations run adaptively adjust the biases more frequently. In later phases 2 and 3, which are initiated with biases optimized in prior phases, longer iterations are helpful to obtain more sampling allowing more refined adjustment to the biases. With the optimized biases, six independent trajectories of 20 ns long production simulations are run, which are merged to generate three trajectory segments, representing 40 ns of sampling each. A combined estimate of the average free energy and standard deviation is obtained from the production simulations (see the [Methods](#) section). We note that on average, the statistical precision of each prediction was ± 0.48 kcal/mol, suggesting our proposed protocol with the sampling regimen noted above are of sufficient precision for prioritizing compounds in lead optimization.

Data Sets. The validation data set was collected from publicly available congeneric ligand sets with experimentally measured binding affinities and a co-crystal structure of the protein in complex with one of the ligands, or a similar ligand in the PDB.¹⁸ Additionally, the choice of the data sets and

ligands within each was motivated by the desire to validate the presented method for its domain of applicability, which is to efficiently screen multiple modifications to a common core compound at single or multiple sites. Seven congeneric sets were collected from various sources, as listed in [Table 1](#). Since most data sets that we found were originally collated to validate FEP approaches, a subset of ligands from each was selected that fit the domain of applicability. A few ligands from each set could not be included because those involved modifications at many different parts of the core and were outside the focus of our current study. Another consideration for selecting the ligands used for validation arose from prior MSLD studies that have found efficient sampling among substituents that were similar in shape or volume.¹² While this consideration was taken into account, the selected ligands were also chosen to model chemically and structurally diverse substituents ([Figures 1–8](#)). The range of affinities for most sets is around 4 kcal/mol ([Table 1](#)), which represents a 1000-fold change in binding affinity, and thus provides an opportunity to test the method's ability to capture large changes in affinity. Four of the seven sets provided an opportunity to test multisite substitutions, which included the proteins MCL1, HSP90, P38 MAP kinase (P38MK), and TNKS2. TNKS2 ligands involving a change in net charge (eight series) were not included. Net charge changes among the ligand series could potentially be modeled accurately in MSLD by applying end-point corrections as done in the context of FEP.³⁰ However, this straightforward application of MSLD was not the focus of the present study. Thirty two ligands from the “left-side” group in [ref 31](#) were included from the fatty acid amide hydrolase (FAAH) data set. Most of the “right-side” ligands involved experimental measurements from racemic mixtures, and thus this set was not included in our study. A subset of ligands from the HSP90 data set from D3R was selected that could be modeled with substituents on two sites. A complete list of ligands examined in the study can be found in the data included in the [Supporting Information](#) files, which includes the ligand and protein structures used to initiate the MSLD calculations.

MSLD Calculations on Diverse Subsets. The seven data sets described above were divided into subsets that differed in number of sites, number of substituents, and chemical and topological diversity of the substituents ([Table 2](#)). This was carried out to evaluate the MSLD workflow to accurately calculate the relative binding free energies under different scenarios. For example, the efficient and reliable sampling of diverse substituents within a single series represents a stringent test of our MSLD workflow implementation. In addition to the differences in ligands, the nature of the protein pocket may also influence sampling and transition rates between different bound states. Thus, 25 data sets listed in [Table 2](#) were constructed to test the protocol adopted in the bias optimization and production stages. Overall, the subsets model 5–32 ligands each using either one or two sites, as shown in the figures below. The 25 subsets categorized by the protein target are discussed below.

MCL1 Data Set. 39 ligands included from this data set were modeled as 5 MSLD systems or subsets which are listed in [Table 2](#). [Figure 1a,b](#) illustrates the ligands modeled in this data set. Systems MCL1-1 to 3 shown in [Figure 1a](#) involve single-site representatives and systems MCL1-4 and 5 shown in [Figure 1b](#) are modeled with two sites each. Systems MCL1-1, 2, 3 include 7, 7, and 6 substituents, respectively, and

Table 2. Subsets Representing MSLD Systems Modeled to Cover Each Validation Set^a

subset ID	N sites	N subs	N exp	range	AUE
MCL1-1	1	7	7	3.18	0.87
MCL1-2	1	7	7	2.70	0.73
MCL1-3	1	6	6	2.56	1.11
MCL1-4	2	4 × 5	13	3.85	0.92
MCL1-5	2	4 × 6	14	3.38	1.16
FAAH-1	1	8	8	4.50	0.92
FAAH-2	1	8	8	4.28	1.00
FAAH-3	1	8	8	4.47	0.78
FAAH-4	1	6	6	2.06	0.74
FAAH-5	1	6	6	1.68	0.63
HSP90-1	2	6 × 5	9	4.30	1.04
HSP90-2	2	8 × 2	9	4.51	1.15
PTP1B-1	1	6	6	1.27	0.55
PTP1B-2	1	6	6	1.74	0.22
PTP1B-3	1	5	5	1.36	0.52
P38MK-1	1	5	5	2.40	0.90
P38MK-2	1	5	5	1.40	1.15
P38MK-3	1	5	5	1.21	0.68
P38MK-4	1	7	7	2.45	1.50
P38MK-5	1	5	5	1.71	1.00
P38MK-6	1	5	5	1.36	1.39
CDK2-1	1	9	9	2.74	0.69
CDK2-2	1	8	8	3.42	0.80
TNKS2-1	2	4 × 8	12	3.53	1.16
TNKS2-2	2	4 × 7	11	4.13	1.12
				average	0.91
				std. dev.	0.29

^aThe subset ID, number of sites (*N* sites), number of substituents (*N* subs), number of ligands with experimental data (*N* exp), experimental affinity range (range), and AUE obtained for each system are listed. Average and standard deviation of AUE are also listed. Units of range and AUE are kcal/mol.

collectively model 18 different aromatic R-groups. The aromatic groups modeled in each system are quite diverse, as a result of which the number of additional heavy atoms in each substituent compared to the common core are 3–7, 3–9, and 4–10 heavy atoms in systems MCL1-1, 2, and 3, respectively. The MSLD simulations involve transitions between the different substituents and thus model changes involving an even larger number heavy atoms. For example, in MCL1-1 system, a transition from substituent 2–3 involves a change in 14 heavy atoms, 7 from substituent 2 (disappearing), and 7 from substituent 3 (appearing). The low AUE obtained for these subsets (Table 2) illustrates the ability of the developed workflow to sample large and diverse groups.

Figure 1b shows two-site systems MCL1-4 and MCL1-5, which involved 4 × 5 and 4 × 6 substituents, respectively. While the two two-site systems modeled 20 and 24 ligands, respectively, only 13/20 and 14/24 substituent combinations had experimental data associated with them. The four substituents involved in site 1 in both these systems are the same. In system MCL1-4, site 2 groups differ in the placement of the chloro and methyl groups, whereas in MCL1-5, the carboxylic acid group involves more diverse chemistry that includes indole, benzofuran, and benzothiophene groups. In MCL1-4, 7 out of the 12 heavy atoms of the heterocyclic carboxylic acid group are included in the MCS, whereas for MCL1-5, due to the diversity of substituents in site 2, only 4

out of 12 heavy atoms are in the MCS. The MCL1-5 system exemplifies the ability of the method to model chemically diverse groups to a greater degree than MCL1-4.

To further examine the efficiency of the landscape flattening we monitored the number of transitions involving each substituent in all 25 subsets, for a total of 185 substituents, this is shown in Figure S1 of the Supporting Information. The first 39 data points correspond to the substituents modeled in systems MCL1-1 to 5. Each point is an average over number of transitions calculated in the three 40 ns trajectory segments. The length of each arm of the error bar equals the standard deviation over the three values. A general observation is that the number of transitions in the protein–ligand complex state are lower than in the free ligand state. However, for most substituents the number of transitions in both states is high. For the MCL1 systems, the number of transitions per 40 ns trajectory segment per substituent range from an average of 220–3541 in the complex state and even higher in the free ligand state. Thus, the data strongly suggest that all substituents are sampled well.

Ligand 27 was common to all the MCL1 subsets, based on which the absolute binding affinities of the entire congeneric set could be calculated from the relative values obtained from each subset. Figure 2a displays the calculated versus experimental absolute binding free energies, ΔG_b , for the MCL1 set. Error bars depict the associated standard deviation.

FAAH Data Set. 32 inhibitors of the FAAH protein previously modeled in a FEP study³¹ were selected. The 32 ligands were divided into five subsets from which single-site MSLD ligand systems were created. MSLD systems FAAH-1, 2, and 3 modeled eight ligands each, whereas FAAH-4 and 5 modeled six ligands each. Ligand 31 was included as a reference in each system. Figure 3a (FAAH-1 to 3) and b (FAAH-4 and 5) illustrate the MSLD systems modeled. The ligands present different R-groups in the ortho, meta, and para positions on the “left” phenyl ring³¹ of the core ligand. Ligands in the set involve differences of up to six heavy atoms with respect to the unsubstituted phenyl ring. Table 2 shows the range of experimental affinities in the five subsets to be 1.68–4.5 kcal/mol, with the range being greater than 4 kcal/mol for three sets. For 31 of the 32 substituents overall, the average number of transitions per 40 ns trajectory segment in the complex state are greater than 70 (Figure S1). The lowest number of transitions were recorded for FAAH-3 sub-2 in the complex state, which models the cyano group at the para position. However, this does not affect the accuracy of the prediction—the unsigned error for this prediction is only 0.14 kcal/mol. For the free ligand state, all substituents involved more than 80 transitions.

The AUE for the five subsets range from 0.63 to 1 kcal/mol, which suggests good predictive power for sets spanning a wide range of affinity. Using ligand 31 as a common reference present in all subsets, absolute binding free energies, ΔG_b , was calculated for all 32 ligands. Figure 2b illustrates the calculated versus experimental ΔG_b , with an overall AUE of 0.7 kcal/mol.

Hsp90 Data Set. Two 2-site systems were modeled to include the HSP90 ligands. Figure 4a,b shows the two MSLD systems. In HSP90-1, site 1 involves R-group decorations on a phenyl ring, whereas site 2 includes more diverse R-groups. In HSP90-2, site 2 contains only two substituents, but site 1 includes substituents that have significant differences in size, and chemistry. For example, substituent site 1-sub 8 includes 11 additional heavy atoms compared to sub 1. In addition, site

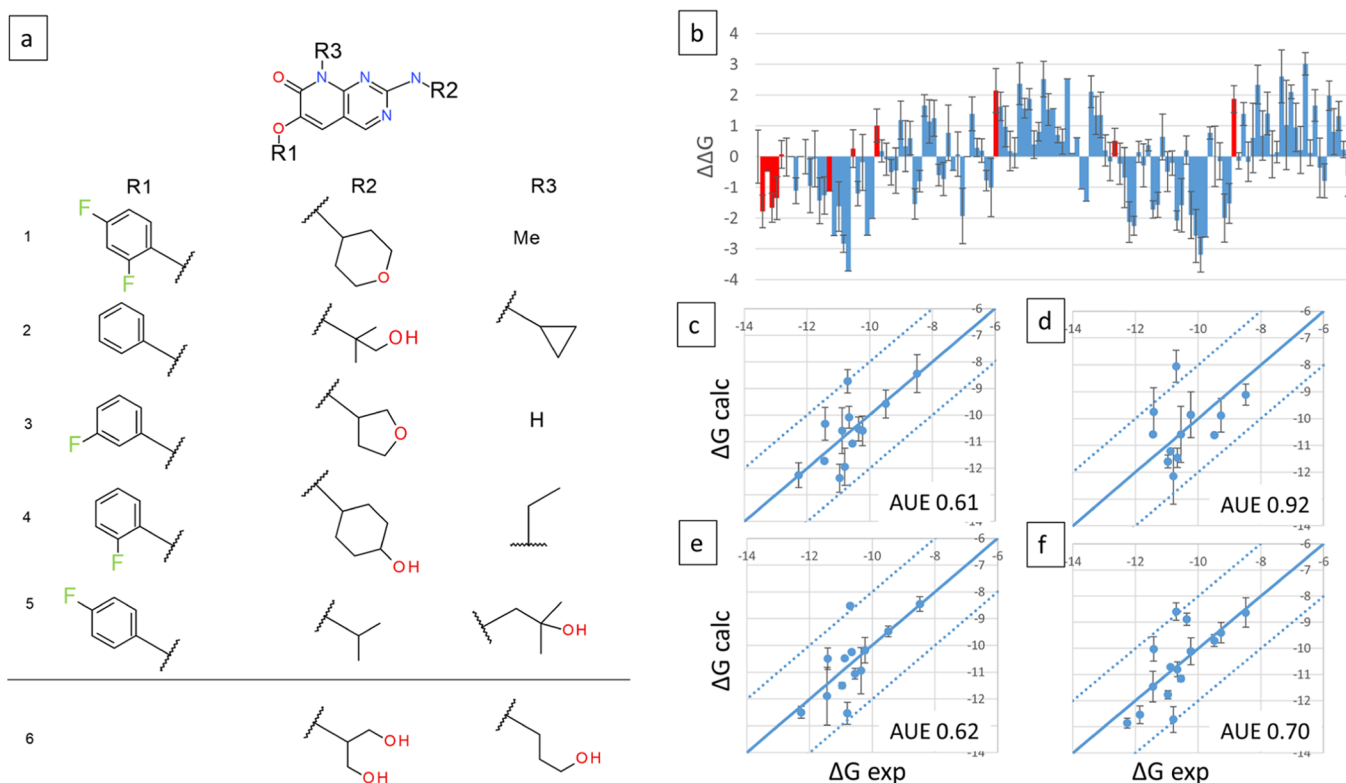


Figure 9. Three-site MSLD systems P38MK-7 and P38MK-8. Panel (a) shows the core ligand with three sites of substitutions and the corresponding substituents. All substituents involved in P38MK-7 are depicted above the horizontal line. One additional substituent each for site 2 and 3 that are modeled in P38MK-8 is depicted below the line. Panel (b) displays the calculated $\Delta \Delta G$ values for 122 out of 125 combinatorial ligands modeled in system P38MK-7, with red bars indicating ligands for which experimental data are available. Panels (c,d) display the calculated vs experimental absolute binding free energies for P38MK-7 and P38MK-8 sets, respectively. Panels (e,f) display binding free energies calculated using the additive approximation for P38MK-7 and P38MK-8 sets, respectively. Calculated relative free energies were converted to absolute by adding a constant offset value that minimizes errors associated with the choice of a single reference ligand, as described in the [Methods](#) section. AUE are indicated on bottom right. Units are kcal/mol.

1 is located in a relatively occluded region of the binding pocket (Figure 4b, inset), such that the transitions from small substituents (e.g., site 1-sub 1) to large ones such as sub8 shown in the figure would require the protein pocket to open up further (discussed below).

Since the two MSLD systems model 6×5 and 8×2 substituents, 30 and 16 free-energy values, respectively, are calculated from the simulations, out of which experimental data exists for nine combinations in each set. The AUE obtained for the two sets with ranges of 4.3 and 4.51, are 1.04 and 1.15 kcal/mol, respectively. The slightly higher error for HSP90-2 system is consistent with the larger size differences among the 8 substituents in site 1. This is reflected in the number of transitions observed for the substituents in the complex state. While substituents 4–8 record more than 40 average transitions, they are relatively low at 5.3, 14, and 4.3 for substituents 1, 2, 3, respectively (Figure S1). The absolute error in predicted ΔG for the ligands that correspond to site 1-sub 1 and site 1-sub 3 are high at 2.22 and 2.24 kcal/mol, respectively, which appears to be consistent with the low number of transitions.

Figure 2c illustrates the calculated versus experimental absolute binding free energies, ΔG_b , for 16 ligands which represent the combinations of substituents for which experimental affinities were available. The AUE for this data set is 0.82 kcal/mol.

PTP1B Data Set. Three single-site MSLD systems were constructed to model ligands from the PTP1B data set with six, six, and five substituents, respectively, as shown in Figure 5. Substituents in all three groups of ligands are not only chemically but also topologically diverse. While Table 2 shows the AUE obtained for the three sets is very low, the experimental binding affinity ranges of the individual sets at 1.27–1.74 kcal/mol is relatively small. However, the accurate results obtained for the set of ligands with R-groups that have significant topological differences shows the power of the method in exploring chemical space broadly. Except for one substituent with 81 average transitions, all have more than 150 in the complex state. The number of transitions in the free ligand state are much higher (Figure S1). Figure 2d plots the calculated versus experimental ΔG_b for the 15 ligands in this data set, with an AUE of 0.39 kcal/mol.

P38 MAP Kinase (P38MK) Data Set. The 27 ligands from the P38MK set were modeled by dividing them into six subsets, which are shown in Figure 6a–c. This data set includes three sites of modifications. The first site is modeled by systems P38MK-1 and 2, the second site is modeled using P38MK-3 and 4, and the third one is modeled using P38MK-5 and 6. The P38MK-1 system models relatively small modifications that involve fluorine substitutions on the phenyl ring. In the P38MK-2 system on the other hand, the linker atom is different or absent (sub 4), resulting in this set being very diverse. Similarly, P38MK-3 to 6 also model groups that

Table 3. MSLD Systems Representing Large-Scale Combinatorial Screening^a

subset	N sites	N sampled	N exp	range	AUE	AUE (additive)
P38MK-7	5 × 5 × 5	122/125	13	3.78	0.61	0.62
P38MK-8	5 × 6 × 6	131/180	15 (12)	2.96	0.92	0.70
MCL1-6	7 × 4 × 4	109/112	16 (14)	4.19	0.84	0.67
MCL1-7	8 × 8	58/64	23 (21)	3.85	1.11	0.90
HSP90-3	6 × 5 × 3	68/90	9 (5)	1.19	0.39	0.43

^aThe subset ID, number of sites (*N* sites), fraction of ligands sampled (*N* sampled), number of ligands with the experimental data that were sampled (*N* exp), experimental affinity range, and AUE obtained for each system are listed. For *N* exp, the number in parentheses if present indicates the number of ligands with experimental affinity sampled in the MSLD simulations. Units of range and AUE are kcal/mol.

are diverse both chemically and topologically. All 32 substituents modeled across the six systems recorded greater than an average of 50 transitions per 40 ns trajectory segment, except for one with 37, indicating sufficiently sampled substituents. Table 2 shows the AUE obtained for each site. P38MK-4 has a relatively high AUE of 1.5 kcal/mol, but for the remaining P38MK sets, it is similar to other ligand series evaluated in this study. ΔG_b for all of the 27 ligands were calculated using P38MK-1 sub1 ligand as a reference and are displayed in Figure 2e. An AUE over all ligands is 0.75 kcal/mol.

CDK2 Data Set. 16 ligands from the CDK2 set were modeled using two single-site systems, CDK2-1 and CDK2-2, which included nine and eight substituents, respectively (Figure 7). The ligands involved differences in R-group substituents on a terminal phenyl ring on meta or para positions or both. All 17 substituents modeled across the two systems recorded greater than an average of 50 transitions per 40 ns trajectory segment, indicating sufficiently converged results (Figure S1). Low values are obtained for the overall AUE for the 16 ligands (Table 1) and for the two sets separately (Table 2). Figure 2f shows the calculated versus experimental absolute binding free energies.

TNKS2 Data Set. 20 ligands from the TNKS2 set were modeled using two two-site MSLD systems depicted in Figure 8. Site 2 in both systems involves substitutions and combinations of fluorine, chlorine, methyl, and methoxy groups on the quinazolinone group. Site 1 in the TNKS2-1 system involved small changes, whereas in the TNKS2-2 system, large flexible alkyl chains were included. While the two two-site systems modeled 32 and 28 ligands, respectively, only 12/32 and 11/28 substituent combinations had experimental data associated with them. All 23 substituents modeled across the two systems recorded greater than an average of 200 transitions per 40 ns trajectory segment (Figure S1), indicating sufficiently sampled substituents. The AUE obtained for the two sets is higher than the overall average of the 25 subsets in this study, but the calculated affinities are still predictive. Figure 2g plots the calculated versus experimental ΔG_b , with an AUE of 0.86 kcal/mol (Table 1).

Screening Large Combinatorial Libraries. The above-described MSLD systems involved one or two sites that incorporated 5–32 combinatorial ligands each. Given the multisite nature of the method, it is of interest to evaluate scaling to larger combinatorial libraries. With this goal in mind, five MSLD systems were created that spanned the same data sets as discussed above. A first and natural choice for a three-site MSLD system was the P38MK data set which includes modifications at three well-separated sites on the core ligand. The three-site system P38MK-7 was constructed to include 5 × 5 × 5 substituents, which is shown in Figure 9a. Table 3 lists

several characteristics and results for the different MSLD systems modeled to predict binding affinities of relatively large combinatorial libraries. For P38MK-7, 122 out of the 125 combinatorial ligands were sampled during the production simulations, predictions for which are displayed in Figure 9b. The $\Delta\Delta G$ values are calculated relative to a reference compound that models substituent 1 on all three sites. About one-third of the ligands are predicted to be as potent as the reference compound based on the error bars overlapping with the $\Delta\Delta G = 0$ line. The remaining compounds exhibit significantly different affinities. This analysis suggests that the calculated affinities of all combinations can help screen for compounds with significantly improved binding affinity. Furthermore, an analysis of the substituents involved in the tightest binding ligands can result in design insights relevant to lead optimization. For example, the top six ligands ranked by favorable affinity include sub 4 at site 2, suggesting its importance in binding. The Discussion section mentioned below illustrates this by presenting a structural analysis of a ligand with substituent 4 on site 2. Experimental data were available for 13 combinations (red bars in Figure 9b), spanning a range of 3.78 kcal/mol. The AUE obtained for this subset of data was 0.61 kcal/mol. Figure 9c plots the calculated versus experimental absolute binding free energies, ΔG_b , for 13 ligands which represent the combinations of substituents for which experimental affinities were available.

To monitor the extent of scalability, a second system, P38MK-8, with the same substituents as in P38MK-7 but with a sixth extra substituent each for sites 2 and 3 was also constructed (5 × 6 × 6 substituents; Figure 9a). For this system, 131/180 ligands were sampled, which suggests that for this system and substituent combination, 180 ligands represent an upper limit of the combinations that can be visited during 120 ns of cumulative sampling. Out of the 49 ligands that were not sampled, 31 ligands involved only substituents 1–5 on all sites. Since nearly all ligands involving these substituents were sampled in P38MK-7 system, the reason for the lesser number of sampled ligands is likely due to the sheer number of combinations and not due to the chemical nature of the additional substituent 6 on sites 2 and 3. The experimental data existed for 15/180 ligands, out of which 12 ligands were sampled. Figure 9d plots the calculated versus experimental ΔG_b . An AUE of 0.92 kcal/mol was observed for these 12 ligands.

For multisite systems covering large combinatorial libraries like P38MK-8, incomplete sampling of all combinatorial ligands arises from the sheer number of ligands which are counted only when all three sites have the corresponding lambda values greater than the cutoff of $\lambda_{\text{cut}} = 0.99$.

The free-energy landscapes of substituent transitions in individual sites are well flattened as evidenced by the number

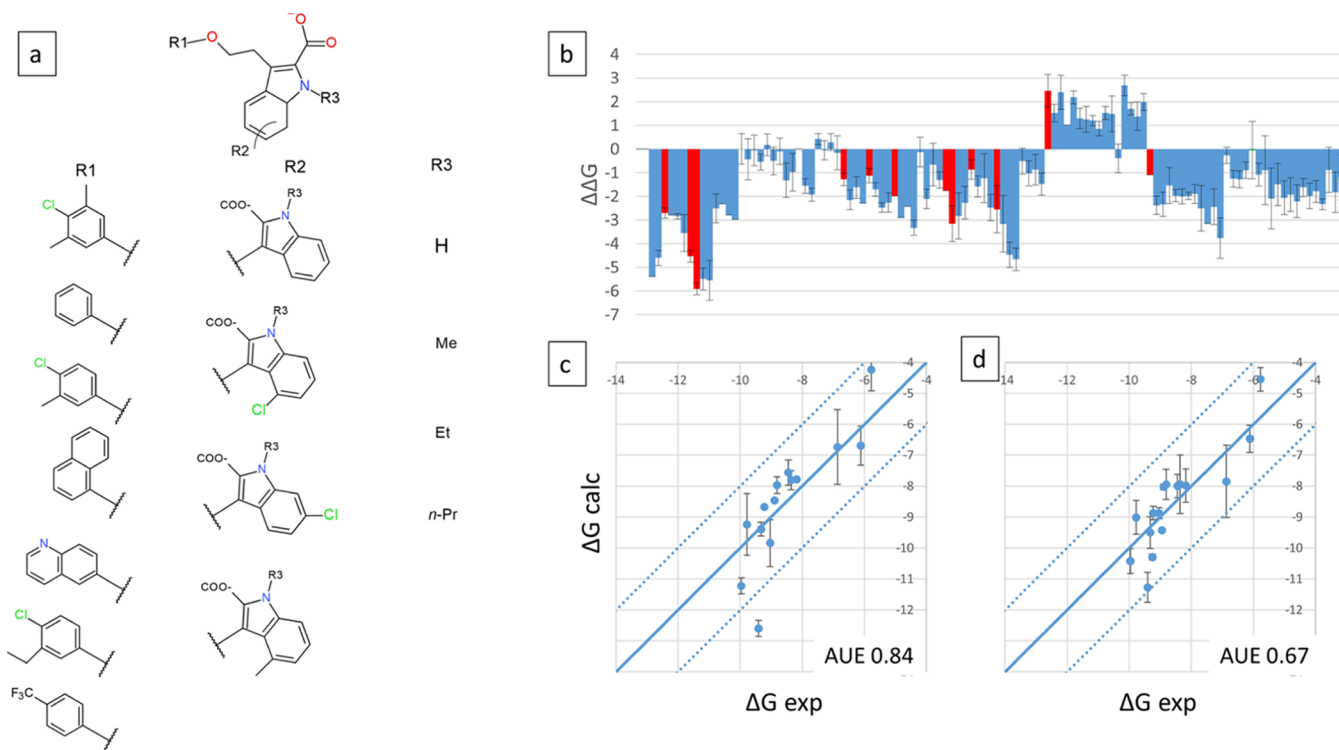


Figure 10. Three-site MSLD system MCL1-6. Panel (a) shows the core ligand with three sites of substitutions or modifications, and the corresponding substituents or modifications. Panel (b) displays the calculated $\Delta\Delta G$ values for 109 out of 112 combinatorial ligands, with red bars indicating ligands for which experimental data are available. Panel (c) displays the calculated vs experimental absolute binding free energies. Panel (d) displays binding free energies calculated using the additive approximation. Calculated relative free energies were converted to absolute by adding a constant offset value that minimizes errors associated with the choice of a single reference ligand, as described in the *Methods* section. AUE are indicated on bottom right. Units are kcal/mol.

of exchanges on each site, which are in a similar range as for the one and two site systems discussed above (data not shown). It is thus of interest to test if a sum of single-site free-energy estimates or “additive estimates,” are predictive, in such contexts where sampling of all combinatorial ligands is incomplete. Additive estimates are obtained by a sum of the single-site free energies corresponding to a combination. While this estimate neglects cooperative effects and may not be accurate when cooperativity is important, it can serve to guide design strategies when the more rigorous cooperative estimates are lacking. For both P38MK-7 and 8, additive estimates were obtained for all 125 and 180 ligands, respectively. Figure 9e,f demonstrates the calculated additive versus experimental ΔG_b . Surprisingly, an AUE of 0.62 and 0.70 kcal/mol were obtained for the two sets, respectively. The implications of additive estimates for this data set and others are discussed below.

Two large-scale MSLD systems were constructed for the MCL1 data set. The first system MCL1-6 modeled $7 \times 4 \times 4$ substituents and is shown in Figure 10a. The core ligand is relatively small, and the data set did not have well-separated three sites of modification. However, for the purpose of testing, the scalability of the presented method, site 2 (in MCL1-4 and 5) was split into two sites, resulting in a three-site system. Additionally, two substituents, ethyl and propyl were added as site 3 and sub 3 and 4, respectively, for which experimental data are not available—these were added simply to test the scalability. 109/112 ligands were sampled during the production simulations, predictions for which are displayed in Figure 10b. The $\Delta\Delta G$ values are calculated relative to a reference compound that models substituent 2, 1, and 1 on

sites 1, 2, and 3, respectively. The predictions show a large number of designs that bind more favorably than the reference compound. Of note is a set of 15 compounds predicted to bind significantly less favorably than the reference compound—all of these compounds model the quinolone group (site 1-sub 5). Such predictions can be useful in deprioritizing design ideas during lead optimization. Figure 10c illustrates the calculated and experimental ΔG for the 14 ligands with the experimental data (indicated by red bars in Figure 10b). Over a range of 4.19 kcal/mol, the AUE is 0.84 kcal/mol (Table 3). An additive estimate of ΔG was obtained for this set and is plotted in Figure 10d, which has an AUE of 0.67 kcal/mol.

A second MCL1 system, MCL1-7, with two sites and 8×8 substituents was constructed, as shown in Figure 11a. Site 1 substituents mostly differ by R-group substituents on the phenyl ring, except for sub 4, which incorporates a naphthyl group, whereas site 2 substituents represent more diverse chemistry. 58/64 ligands were sampled during the production runs, predictions for which are displayed in Figure 11b. The $\Delta\Delta G$ values are calculated relative to a reference compound that models substituent 2, 1 on sites 1, 2, respectively (same compound as the one used in MCL1-6 system). The predictions show a large number of designs that bind significantly more strongly than the reference compound. 21 ligands have experimental data, indicated by red bars in Figure 11b. The non-additive and additive estimates of ΔG were calculated and are plotted in Figure 11c,d, respectively. The experimental data span 3.85 kcal/mol over which the AUE is 1.11 and 0.9 kcal/mol, respectively (Table 3).

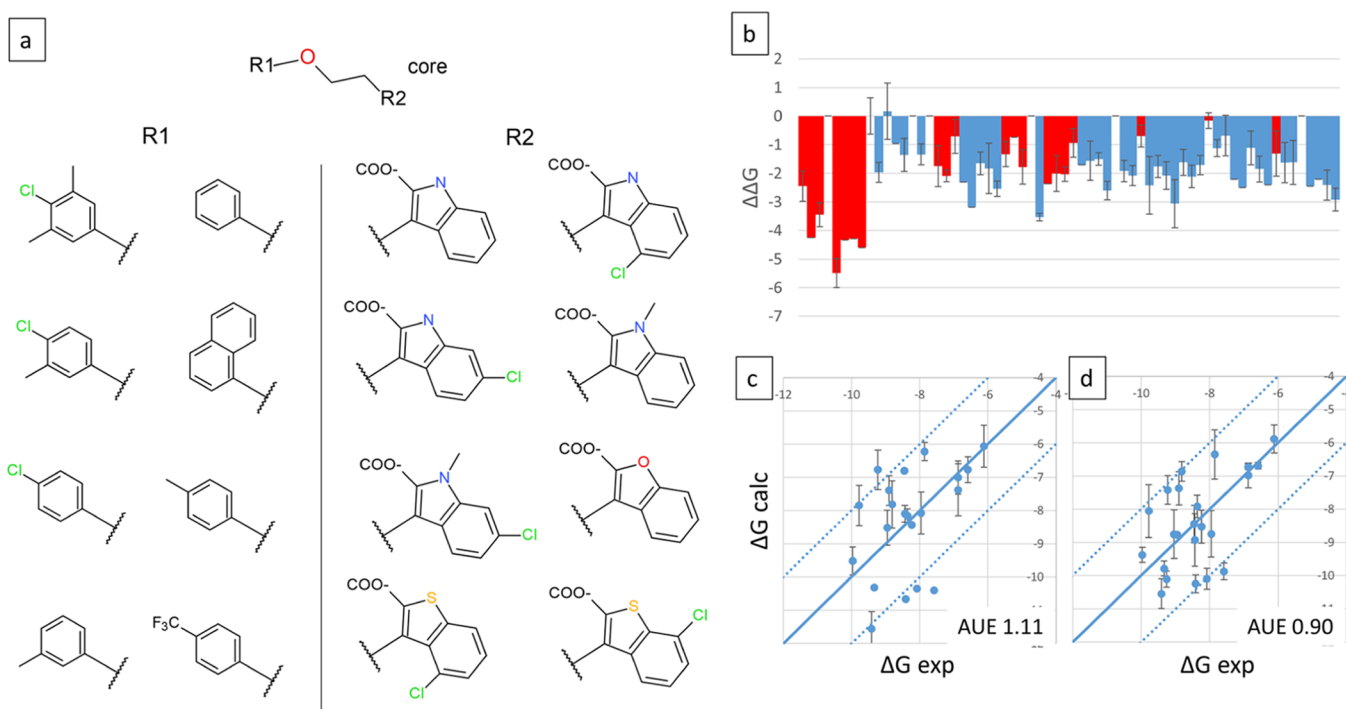


Figure 11. Three-site MSLD system MCL1-7. Panel (a) shows the core ligand with two sites of substitutions and the corresponding substituents. Panel (b) displays the calculated $\Delta\Delta G$ values for 58 out of 64 combinatorial ligands, with red bars indicating ligands for which experimental data are available. Panel (c) displays the calculated vs experimental absolute binding free energies. Panel (d) displays binding free energies calculated using the additive approximation. Calculated relative free energies were converted to absolute by adding a constant offset value that minimizes errors associated with the choice of a single reference ligand, as described in the [Methods](#) section. AUE are indicated on bottom right. Units are kcal/mol.

Finally, a third site was added to the HSP90-1 system to result in the three-site HSP90-3 system with $6 \times 5 \times 3$ substituents, as shown in Figure S2 of the [Supporting Information](#). The methyl and ethyl substituents were added simply to evaluate the scalability; experimental data points for ligands with these substituents do not exist. In this system, only 68/90 ligands were sampled, predictions for which are displayed in [Figure S2b](#). The $\Delta\Delta G$ values are calculated relative to a reference compound that models substituent 5, 1, and 1 on sites 1, 2, and 3, respectively (the ligand in the co-crystal structure 4YKR used in the simulations). The number of sampled ligands is lower than the other systems discussed above, which resulted in sampling a much larger fraction of the ligands modeled. This observation suggests that factors other than the number of substituents can impact sampling. Analysis of the number of transitions at each site revealed that the number of exchanges between site 3-sub 1 and sub 2 or 3 are low. More than 1500 transitions are observed between sub 2 and 3 in each 40 ns trajectory segment, but the average number transitions involving sub 1 are much lower at 97. While results for one and two site systems presented above suggest that statistics from such a number of transitions result in accurate predictions, for a three-site system, the large number of combinations becomes prohibitively high. Experimental binding affinities are available for nine ligands, out of which five were sampled, which are shown in [Figure S2c](#). [Figure S2d](#) displays the additive estimates of free energies. Both estimates result in accurate predictions ([Table 3](#)), but the additive estimate is able to provide predictions for all 90 combinatorial ligands.

DISCUSSION

Given an accurate force field, explicit solvent-based relative free-energy methods can potentially offer accurate protein–ligand binding affinity predictions by virtue of an atomically detailed representation and a rigorous statistical thermodynamic formulation. Both of these virtues are absent from several alternative methods for estimating binding affinity, including docking–scoring methods, and implicit solvent-based end-point methods, including MM–GBSA and MM–PBSA.^{34,35} Approximate methods are useful in prioritizing compounds from large sets; however, the approximations employed have an unpredictable effect on the accuracy spanning different data sets and therefore render the methods less reliable. In contrast, free-energy methods are expected to be reliable based on the above cited considerations. In the recent past, several retrospective^{4–6,36,37} and prospective^{31,33,38} studies have shown FEP to perform reliably for a large number of data sets and projects. However, relative FEP relies on $\Delta\Delta G$ estimates for pairs of ligands, which are obtained from calculations requiring tens of nanoseconds of sampling each, thus making the screening of large libraries of ligands computationally expensive. MSLD offers a fundamentally different and efficient solution for the calculation of relative binding free energies.

Applicability Domain. Prior applications of MSLD have tested its applicability in estimating binding free energies of a chemical series that model both small modifications¹⁵ and substantially larger ones.¹⁶ In this work, we chose an applicability domain motivated by the fact that MSLD simulations involve efficient transitions between substituents that are similar in volume or shape.¹² To illustrate this

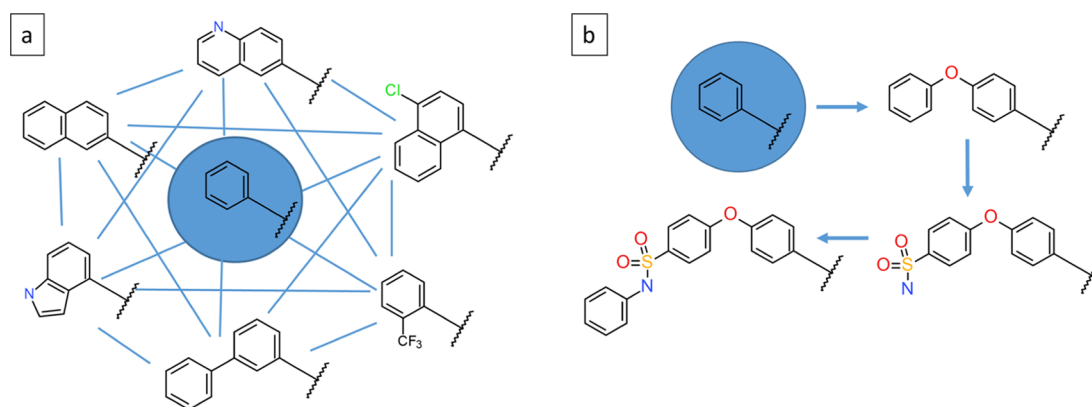


Figure 12. Example illustrating the domain of applicability of the presented MSLD workflow. Panel (a) shows a representative example with a core substituent and a collection of substituents to explore chemical space in the neighborhood of the core. Blue lines indicate potential transitions during the MSLD simulations, which represent a fully connected network. Panel (b) shows an example of a series of ligands beyond the scope of the ligand sets demonstrated in this study. Arrows indicate a possible relative FEP calculation plan to cover this data set, which represents a minimally connected FEP map.

applicability domain, Figure 12a shows a representative screening scenario using ligands modeled in the MCL1 data set supported by the presented validation, where different modifications to a core compound (circled in center) are modeled. MSLD simulations allow for all-to-all transitions (blue lines) and thus model a fully connected network. Figure 12b shows a contrasting example of a series of ligands beyond the scope of the ligand sets demonstrated in this study, in which the modifications involve larger changes (up to 16 heavy atoms) within the series. Blue arrows represent a minimal set of FEP calculations that allow the determination of affinities for this set. In principle, such large changes could still be modeled with MSLD, but longer simulations than that used in this study may be needed to obtain accurate estimates. The applicability of the presented workflow to such cases will be the subject for a future study. Importantly, the hypothetical example shown in Figure 12a and the real data sets modeled in this study (Figures 1 and 3–11) show that the method can still allow for the modeling of diverse substituents and modifications. For example, while R-group substituents were modeled in FAAH-2 (Figure 3) and CDK2-1 (Figure 7) systems, different heteroaromatic groups were included in MCL1-1 (Figure 1) and HSP90-2 (Figure 4) systems. The PTP1B-1 system (Figure 5) is an example of modeling differently sized rings. Different flexible alkyl chains were modeled in P38MK-4 (Figure 6) and TNKS2-2 (Figure 8) systems. These examples show the capability of the method to handle large differences in chemistry and topology and therefore can be used to explore chemical space broadly.

Second, it is expected that a larger number of transitions and therefore better statistics would be related to a smaller number of substituents. Thus, one has to balance the desire to screen large libraries with the quality of statistics obtained. Figure S3 and the accompanying text in the Supporting Information present a detailed analysis of the number of transitions and its dependence on the number of substituents. The accurate predictions obtained with eight or nine substituent systems modeled in this study (e.g., FAAH and CDK2) suggest that such a large number of substituents can be modeled accurately. We also note that eight or nine substituents have been identified in prior studies as an upper limit of the number of substituents that can be reliably modeled on a site.^{14,15} A larger number of substituents can result in poorer statistics and lower

confidence in the predictions. Finally, another requirement associated with the implemented method is that the sites should not have an overlap of atoms between them and ideally be separated by a few bonds. Separation of a few bonds is required to avoid many intersite shared valence parameters that may lead to instability during the simulations.

Additive Estimates. An interesting result for systems modeling large combinatorial libraries (Table 3) is that additive free-energy estimates were marginally more accurate than the cooperative estimates. This is somewhat surprising because additive estimates neglect the cooperative binding contributions from different sites. A possible reason for this could be better statistics for the additive estimates involving a larger number of substituent exchanges—for example, for the three-site MCL1-6 system, the average number of single-site transitions for the three sites are 864.8, 3431.3, and 2159.5 over 40 ns. However, when transitions are calculated for the simultaneous occupancy of three sites, the corresponding average number is 48.9 (data not shown). It is anticipated that for systems that have strong intersite coupling, the additive approximation will become less accurate, even though it is not apparent in any of the systems tested here. This too may be the result of the design principles used in choosing how to delineate multisite systems, whereby one or more bonds separated the branching of substituents likely reducing potential cooperative effects. Thus, it would be recommended to use the more rigorous cooperative estimates when available and use additive estimates when the former is unavailable due to incomplete sampling. One prior MSLD application has shown scalability on larger combinatorial libraries,¹⁵ but that study involved small substituents, used replica exchange to enhance sampling further, and more importantly, used about an order of magnitude more sampling than used here. In the interest of being of utility in fast paced lead optimization projects, the amount of sampling used in this study is much smaller and yet shows good predictability.

While additive estimates do not explicitly incorporate cooperativity, MSLD by design is likely better suited than FEP to obtain additive estimates for combinatorial libraries. Since MSLD incorporates multiple sites in a single system, the effect of other sites on the exchange statistics of one site are represented in a “mean field” manner. If one were to do an analogous calculation with FEP, only a single site would be

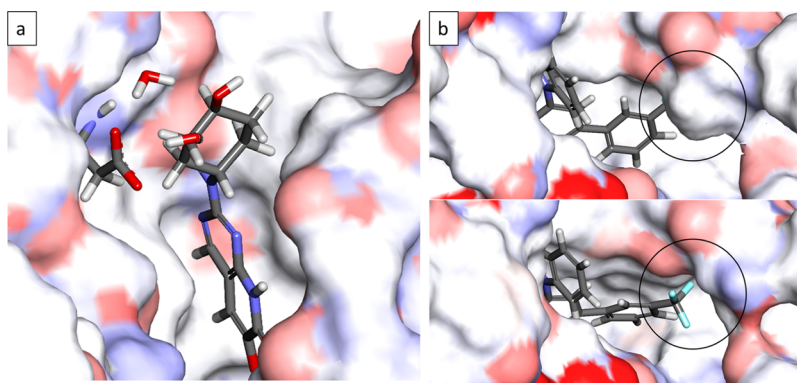


Figure 13. Representative snapshots chosen from MSLD simulation trajectories. Panel (a) shows the ligand with the cyclohexanol group (site 2-sub 4) from the P38MK-7 system forming a water-mediated hydrogen bond with the protein. The snapshot coincides with the lambda value of the substituent depicted being close to unity. Panel (b) shows the tri-fluoro phenyl group (site 1-sub 8) from HSP90-2 system. Top sub-panel shows a snapshot corresponding to the lambda value of site 1-sub 8 being close to zero, and the bottom sub-panel corresponds to the group interacting with the protein with a lambda value close to unity. The circled region shows a change in the protein conformation to accommodate the tri-fluoro group.

evaluated in each simulation, while fixing the other sites to an arbitrarily chosen substituent. The inclusion of multiple sites in a single system is a better approach in this scenario and is a likely reason for obtaining accurate estimates in the five systems tested here (Table 3). Finally, it is noteworthy that the success is seen here for the single-site additivity approximation combined with the large chemical spaces it opens to study, suggesting that the application of such multisite calculations would be of significant utility in establishing QSARs for a targeted scaffold.

Estimates Obtained from Reduced Sampling. Results presented above show the method's efficiency in screening large compound sets. However, it is still of interest to test whether shorter simulation times could result in comparable accuracy as that would enable screening of even larger libraries or number of systems, given limited computing resources. To test the effect of sampling duration of MSLD production runs, the total sampling time was halved from 120 to 60 ns. Figure S4 and Table S1 in the Supporting Information show the AUE obtained for the seven subsets and the five large combinatorial libraries, respectively. Results show only an insignificant drop in accuracy with reduced production sampling for the seven subsets. For large combinatorial libraries, reduced sampling results in lesser number of ligands being sampled, but the additive free-energy estimates (Table S1) are nearly as accurate as those obtained with 120 ns of sampling (Table 3). Thus, it may be possible to obtain accurate estimates with reduced sampling.

Another approach to gain efficiency was explored by obtaining free-energy estimates from fixed biases provided by phase 1, 2, and 3 of the bias optimization stages. The fixed bias estimates that are obtained from the bias optimization protocol estimate the end-state free energy, and together with the additive approximation, can predict the binding free energies. Figures S4 and S5 in the Supporting Information present the AUE obtained for the seven protein sets and the five large combinatorial libraries, respectively. These results show that binding free energies obtained from fixed bias estimates have significantly reduced accuracy compared to those obtained from production simulations, and therefore, the production stage is highly recommended for improved accuracy. Finally, the necessity of a three-phase bias optimization was examined by performing 120 ns of cumulative production sampling but initiated from phase 1 and phase 2 of the bias optimization

stages. The analysis of the prediction metrics are detailed in the Supporting Information, with Tables S2–S4 listing the accuracy obtained with different levels of bias optimization. Results show that while for many sets, lesser bias optimization does not affect accuracy, for a few cases, it leads to lack of sampling for a portion of the ligands (P38MK-7 and P38MK-8 systems). Given an additional computational cost per phase of only 10 ns, it is recommended that the three-phase protocol be used for reliable performance.

Structural Insights Obtained from MSLD Simulations.

In addition to binding free-energy predictions, MSLD simulation trajectories can provide structural insights into the binding of different ligands modeled. Such information could be useful in generating additional design ideas that may result in improved affinity. For example, the site 2-sub 4 group in P38MK-7 system is present among the six topmost favorable compounds, and it may be of importance to understand the interactions leading to the favorable affinity of ligands that include this substituent. Production simulation trajectories of the P38MK-7 system were analyzed visually to identify frames that coincide with the lambda value of site 2-sub 4 being close to unity. This analysis revealed a water-mediated interaction between the cyclohexanol alcohol group in site 2-sub 4 and the carboxylate group of ASP109 in the protein. Figure 13a shows a representative snapshot displaying this water-mediated interaction, where one of the two water molecules forms a water-mediated hydrogen bond between the ligand and the protein.

Another example of the structural and dynamical information gleaned from the simulations is from the HSP90-2 system, where the interaction of the site 1-sub 8 group leads to widening of the protein pocket. The top panel in Figure 13b shows a representative frame corresponding to site 1-sub 8 being non-interactive, with its lambda value close to zero, where the tri-fluoro group clashes with the protein. The bottom panel shows a representative frame when its value is close to unity. The encircled region in the panels show a significant change in the shape of the protein pocket which is a consequence of the breaking an intra-protein hydrogen bond between THR94 sidechain and ASN91 backbone. This information can direct the use of relevant protein conformations in subsequent designs involving functionalization of this substituents to further enhance binding affinity.

Sampling of Different Conformational States of Ligands. Accuracy in free-energy calculations requires the sampling of all thermodynamically relevant conformational states. This can be challenging when multiple stable conformational states exist and are separated by high free-energy barriers. One prominent example of alternate conformational states are ortho- and meta-substituted phenyl rings frequently found in drug-like molecules. Typically, phenyl rings interact closely with the protein pocket which often prevents a rotation of the ring in the bound state, which leads to the sampling of only one of the two conformational states during free-energy calculation, potentially leading to inaccurate results. Thus, as a general practice, initial conformations used in free-energy calculations should be selected so as to favor the thermodynamically most favorable state as best as possible, doing which would at least ensure the more stable conformational state being sampled during the calculations. To this end, we used the automated GAC²¹ method implemented in Discovery Studio (and Pipeline Pilot) that detects the different rotameric states automatically and identifies the most favorable using an MM-GBSA score. Thus, the initial conformations used in the free-energy calculations here or in FEP are likely the thermodynamically favorable ones. However, limitations in scoring due to the implicit solvent approximation, or the occurrence of isoenergetic conformations may still limit accuracy. In such cases, MSLD by design offers a better and generalizable solution to sampling the different conformations during the simulations compared to pairwise FEP for two reasons. First, since multiple ligands are incorporated by means of the multitopology ligand, the common core region is smaller than it would be in pairwise FEP, thus allowing for more flexibility for each substituent to explore different conformations. Second, in MSLD for long periods of sampling time during the course of the entire simulation, individual substituents have lambda values close to zero when conformational transitions would be easy to achieve. In contrast, in FEP non-zero lambda values only exist in the end-state windows, thus not allowing for such transitions during the remaining lambda windows. Figure 14 shows two examples of alternate conformational states being sampled during the protein–ligand complex MSLD simulations. Figure 14a shows FAAH-4 system where only the core atoms, and those belonging to substituent 1 are shown. Two phenyl ring orientations sampled during

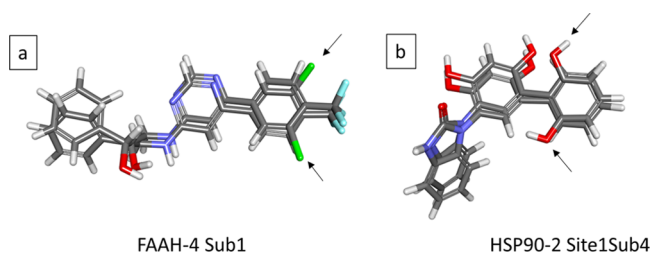


Figure 14. Two examples of alternate conformational states being sampled during the protein–ligand complex MSLD simulations. Panel (a) shows FAAH-4 system, where only the core atoms and those belonging to substituent 1 are shown. Two phenyl ring orientations sampled during different times in the first simulation's trajectory are shown, which position the chlorine atom at the two locations. Panel (b) shows another example, HSP90-2 system, where only the core atoms and those of site 1-substituent 4 are shown. The two snapshots of the ligand show the two orientations of the phenyl ring that position the alcohol group differently.

different times in the first simulation's trajectory are shown, which position the chlorine atom at the two locations. Figure 14b shows another example, HSP90-2 system, where only the core atoms and those of site 1-substituent 4 are shown. The two snapshots of the ligand show the two orientations of the phenyl ring that position the alcohol group differently. We note that not all systems with phenyl ring analogs modeled in this study allow the free rotation like the above two examples. For some systems (e.g., MCL1-1), part of the phenyl ring is included in the common core, which prevents substituents from independently accessing the two rotameric states. The extent of the common core is dependent on the extent of differences modeled and on the partial charge model used. In this study, bond charge increments²⁵ are used to assign partial charges, which typically result in minimizing the differences to a few atoms more than the chemical modifications themselves.

Computing Efficiency. Several features of MSLD make the method more efficiently relative to conventional FEP. First, the dynamic treatment of lambda in combination with optimized biasing potentials enables the full thermodynamic transition between any pair of states to be modeled in a single simulation, where sampling is driven by the intrinsic chemical and conformational free-energy landscape. In most FEP applications reported to date, a predetermined lambda schedule is adopted which is independent of the free-energy landscape and thus not necessarily the most efficient. Second, multiple ligands are modeled in a single simulation. In this study, 19 single-site systems were tested which modeled 5–9 ligands each (Table 2). A cumulative sampling time of 150 ns yielded predictive estimates from these systems. FEP applications that include cycle-closure edges typically use $1.5 \times N$ relative FEP calculations to obtain absolute free-energy estimates of N ligands.⁴ Assuming 60 ns of sampling per relative FEP calculation, it amounts to 450–810 ns of sampling for 5–9 ligands. For single-site MSLD systems used here, predictive estimates could be obtained with 3–5.4 fold less sampling than typical FEP applications. The two-site systems used in this study modeled 16–64 ligands, for which predictive estimates could be obtained with 150 ns of cumulative sampling. For these systems, MSLD required 9.6–38.4 fold less sampling than FEP would have required, based on the above considerations. Finally, for three-site systems, considering the $5 \times 5 \times 5$ P38MK-7 as an example, predictive free energies were obtained with 75-fold less sampling. The error bars in the free-energy plots shown in Figures 2 and 9–11 indicate an acceptable degree of precision that would enable the use of the predictions in prioritizing compounds.

The throughput that can be obtained with the MSLD workflow is also dependent on the speed of the MD engine. As an example, the P38MK protein solvated in an orthorhombic shaped water box has 31,282 atoms. On this system, MSLD sampling of 32 ns can be obtained per day with the current implementation run on one NVIDIA GTX 1080 Ti card along with 4 CPUs. Thus, a typical calculation reported in this study would require 5 GPU days' worth of resources, with minor additional resources for the free ligand system. The bias optimization stage with default parameters required 30 ns or about 1 day on a single GPU. The production calculations running six trajectories on one GPU each require about 15 h to finish 20 ns. Thus, in less than 2 days of wall clock time, a large combinatorial library could be screened with small-scale computing resources. Moreover, if three trajectories instead

of six are chosen to run production calculations, it would reduce the computing resources requirements by 40%.

Current Best Practices. Based on the lessons we learned from the data sets and calculations described in this study, briefly summarized here are the factors to consider when applying the MSLD workflow to new systems. First, we note that the method implemented in its present form is most efficient for screening ligand substituents that span a similar volume or shape in the binding pocket. As shown above, this domain allows for screening chemically and topologically diverse groups. Second, to ensure sufficient sampling, our experience suggests that the number of substituents should not exceed nine at each site of modification. If a larger number of modifications is desired to be screened at a single site, the data set can be split into multiple groups, with one common ligand serving as a reference within each group, as was done in the FAAH and CDK2 data sets, for example. We also note that such data set splits would be straightforward to automate, though it was not carried out in this study. Third, when setting up multisite systems, care must be taken to ensure that the site centers be separated by at least four bonds to prevent overlap between groups from two sites. Fourth, using a cumulative 120 ns of production sampling used in the workflow, about 100 combinatorial ligands can be screened. However within the same frame of sampling, a larger number of ligands can be screened if estimates employing the additivity approximation are utilized. Our experience from these calculations suggest that this will likely be a good approximation in many cases. Finally, given the fact that accuracy was maintained with a reduced cumulative production sampling time of 60 ns, the workflow can be used with reduced sampling to improve throughput, especially screening relatively small libraries or when the additive approximation is used in multisite systems.

CONCLUSIONS

This study presented an automated workflow that provides a framework to set up and execute the MSLD free-energy method to calculate relative binding affinities of combinatorial libraries of ligands. Given the specification of a core ligand with 1–3 well-separated sites and a list of modifications, the workflow sets up simulation systems and executes the calculations. An automated protocol to iteratively optimize biasing potentials enabling enhanced sampling production simulations on a flattened free-energy landscape is retrospectively validated on 31 MSLD systems spanning a wide range of chemical series and proteins. With a cumulative sampling time of 150 ns, estimated binding free energies accurately predict experimental trends for data sets that include structurally diverse substituents. The presented method thus allows for accurate estimation of binding free energies of large combinatorial libraries with small-scale GPU computing resources.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.0c00830>.

Protein structures in MOL2 format, downloaded from PDB and prepared using BIOVIA Discovery Studio and used in the calculations; MOL2 files containing the ligands with assigned force-field parameters and in the conformation used to prepare the 31 MSLD systems

used in validation; and Excel spreadsheet containing calculated free energies (ZIP)

Figures and tables referred to in the paper (PDF)

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Notes

The authors declare the following competing financial interest(s): E. Prabhu Raman is an employee of Dassault Systemes BIOVIA. The Brooks group receives or has received funding from Dassault Systemes BIOVIA in support of world-wide CHARMM development efforts.

The MSLD workflow, as described in this manuscript, is implemented as a set of protocols in BIOVIA Discovery Studio and Pipeline Pilot packages. A prototype version of the functionality is available in the Discovery Studio 2020 release. For further details, including information about availability of the MSLD bias optimization and production protocol described in this study, please see: <https://www.3ds.com/products-services/biovia/products/molecular-modeling-simulation/biovia-discovery-studio/>.

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